دفتر مادة

تفاضل وتكامل ٢

خدمتكم عبادة نتقرب بها الى اللّه

إعداد :مهند سرور



لجنة سنافر البوليتكنك

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سنافر البوليتكنك 2022

لجنة سنافر البوليتكنك

CH: 6

6.1:- Exponential and Logarithimic functions

Def:- if $f(x)=b^x$, $b>0$ and $b\neq 1$
then f is called an exponential function with base b

ex:-

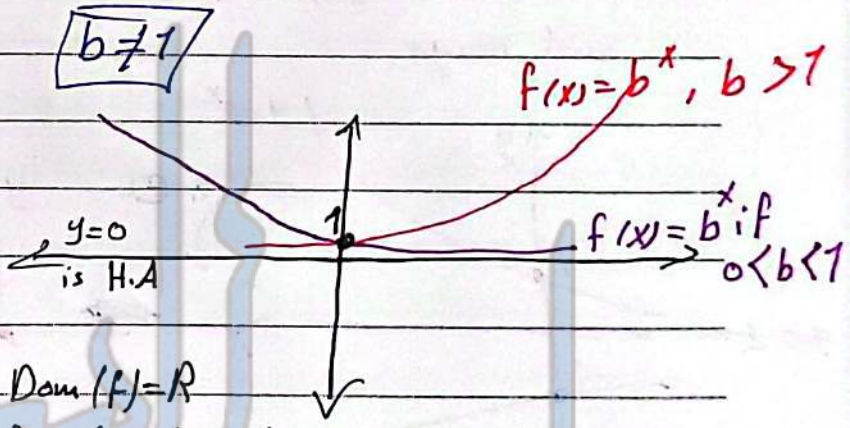
$f(x)=b^x$

$f(x)=1^x=1$ → is constant

$f(x)=(-1)^x$

$f(\frac{1}{2})=(-1)^{\frac{1}{2}}=\sqrt{-1}\notin R$

$b\neq 1$



Dom(f)=R

Rag(f)=(0, ∞)

Laws of exponents: if $a,b>0$ and x,y are real numbers

Then

① $a^{x+y}=a^x.a^y$

② $a^{x-y}=\frac{a^x}{a^y}$

③ $(a^x)^y=a^{xy}$

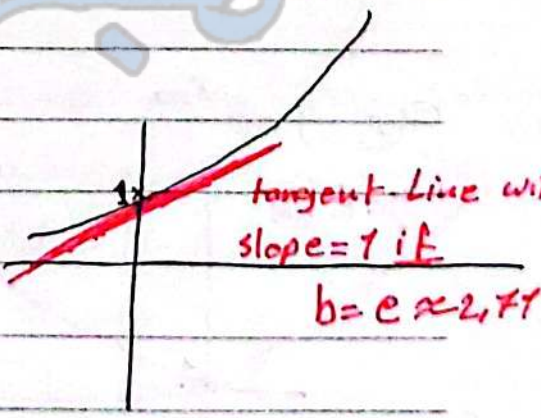
④ $a^{-x}=\frac{1}{a^x}$

⑤ $(a.b)^x=a^x.b^x$

⑥ $(\frac{a}{b})^x=\frac{a^x}{b^x}$

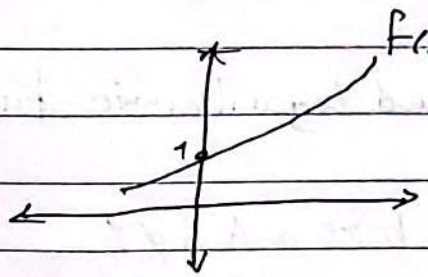
⑦ $a^x=a^y \longrightarrow x=y$

⑧ $a^x<a^y \Rightarrow x<y$



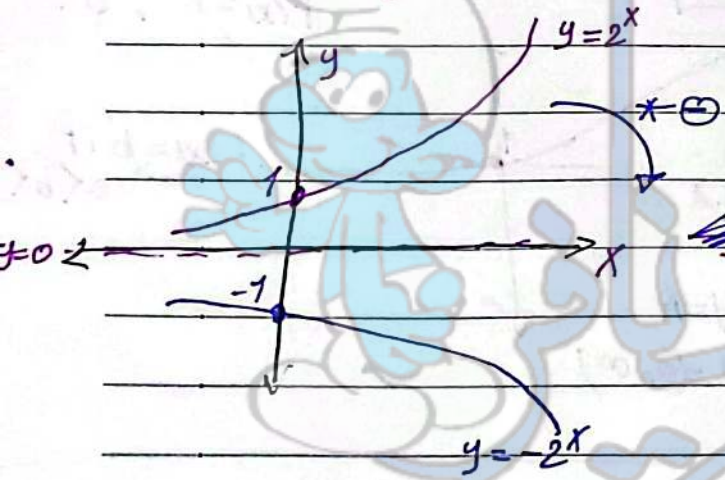
Amazing

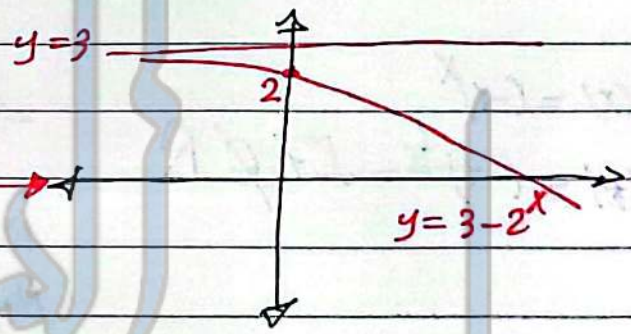
Def: $f(x)=e^x$ is called a natural exp. fun. with base e.



Dom = R and Rang = $(0,\infty)$

ex:- sketch the graph of $y=3-2^x$ and determine its domain and range

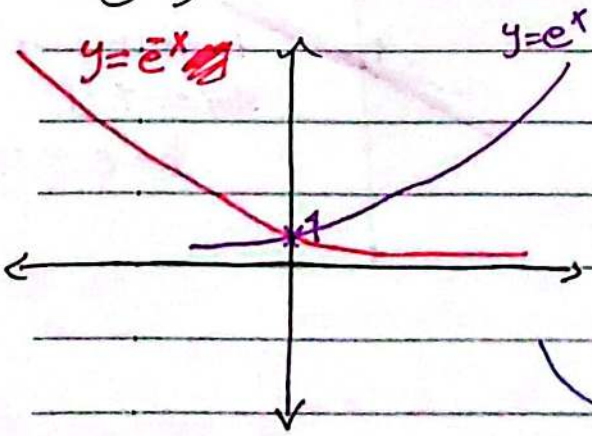


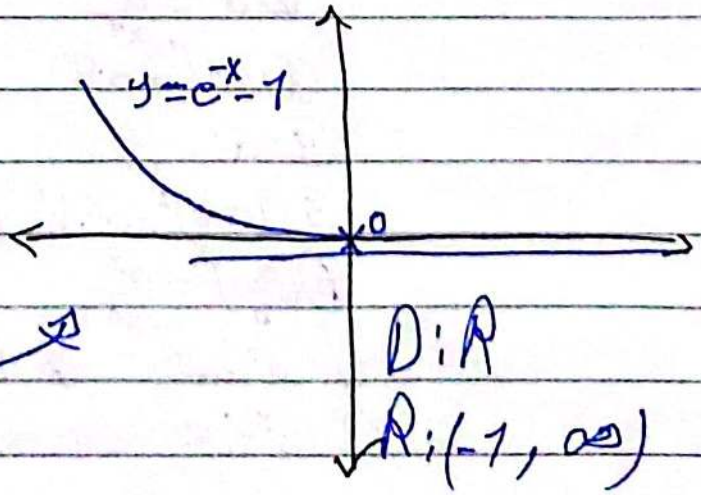


Dom = R

Rang = $(-\infty, 3)$

ex② $y=e^{-x}-1$





D: R

R: $(-1,\infty)$

Amazi

Ex: Solve the following:

① $(64)^x = \frac{1}{16} \rightarrow (2^6)^x = \frac{1}{2^4} \rightarrow \frac{6x}{6} = \frac{-4}{6} \quad x = \frac{-2}{3}$

② $e^{x^2-2x} = 1 \rightarrow e^{x^2-2x} = e^0 \rightarrow x^2-2x=0 \rightarrow (x-2)x$

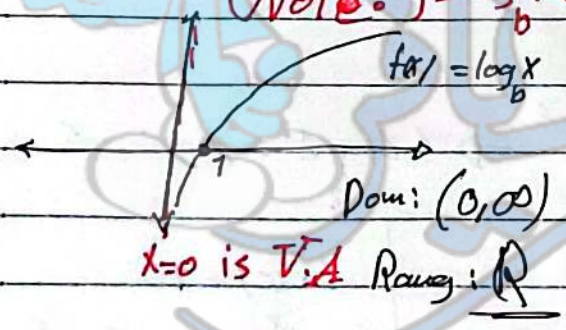
$\rightarrow x=0$

$\rightarrow x=2$

Def: if $f(x) = b^x$, $b>0$ and $b \neq 1$ then

$f^{-1}(x) = \log_b x$ is called Logarithmic function with base b.

(Note: $y = \log_b x \leftrightarrow x = b^y$)



Laws of Logarithms: if x and y are positive numbers then:-

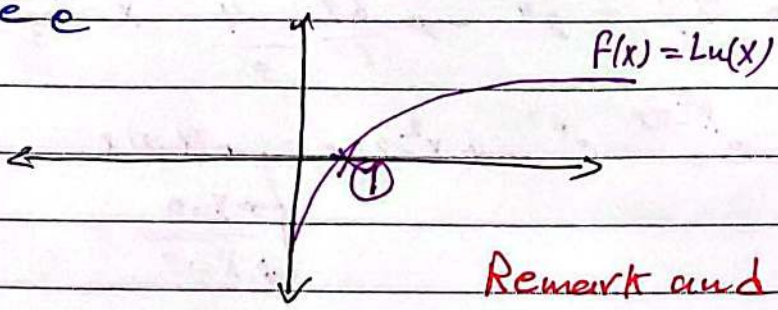
① $\log_b b^x = x$ ② $b^{\log_b x} = x$ ③ $\log_b(xy) = \log_b(x) + \log_b(y)$

④ $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$ ⑤ $\log_b x^r = r * \log_b x$

⑥ $\log_b(1) = 0$ ⑦ $\log_b(b) = 1$ ⑧ $\log_b x = \log_b(y) \Leftrightarrow x = y$

Def : if $b = e = 2,71$ Then

$f(x) = \log_e x = \ln x$ is called a natural Logarithm function with base e



Remark and ⑨ $\log_b X = \frac{\ln X}{\ln b}$

Ex : Evaluate

$\log_2 80 - \log_2 5 + \log_2 4$

$= \log_2 \frac{80}{5} + \log_2 4$

$= \log_2 16 \times 4 \longrightarrow \log_2 64 \Rightarrow \log_2 2^6 \Rightarrow 6 \cdot \log_2 2 = 6 \times 1 = 6$

Exe solve the following

① $\ln x = 5$

Sol : $e^{\ln x} = e^5 \longrightarrow x = e^5$

② $\log_3 (2x-1) = \log_3 (7)$

Sol : $2x - 1 = 7 \Rightarrow 2x = 8 \longrightarrow x = 4$

$\ln(e) = 1$

$\log_b b^x = x \longrightarrow \ln e^x = x$

$\log_b X$

$b = X \longrightarrow e^{\ln x} = X$

Amazing

② $5^x = 9$

Sol: $\log_5 5^x = \log_5 9$

$\Downarrow$

$x = \log_5(9)$

④ $e^{x^2-2x} = 2$

Sol: $\ln e^{x^2-2x} = \ln 2$

$x^2 - 2x - \ln 2 = 0$

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$= \frac{2 \pm \sqrt{4+4\ln 2}}{2}$

⑤ $\log_{10} x = \sqrt{2}$

Sol: $\log_{10} x = \sqrt{2}$

$10^{\log_{10} x} = 10^{\sqrt{2}}$

$x = 10^{\sqrt{2}}$

⑥ $\ln(x-1) + \ln(x) = 1$

Sol: $\ln(x-1)\cdot x = 1$

$= \ln(x^2 - x) = 1$

$e^{\ln(x^2-x)} = e^1$

$x^2 - x = e^1 \rightarrow x^2 - x - e^1 = 0$

$x = \frac{1 \pm \sqrt{1+4e}}{2}$

$x = \frac{1+\sqrt{1+4e}}{2}$ ✓

$x = \frac{1-\sqrt{1+4e}}{2}$ ✗ impossible

Because $\frac{1-\sqrt{1+4e}}{2} < 0$

⑦ $\ln(\ln(x+1)) = 2$

Sol: $e^{\ln(\ln(x+1))} = e^2$

$= \ln(x+1) = e^2$

$e^{\ln(x+1)} = e^{e^2}$

$x+1 = e^{e^2} \rightarrow x = e^{e^2} - 1$

Amazing

(8) $\frac{e^x - e^{-x}}{2} = 1$ $= e^x - e^{-x} = 2$ $\rightarrow \ln e^x - \ln e^{-x} = 2\ln 2$

$= x + x = 2\ln 2$

$(e^x - e^{-x} = 2) \cdot e^x = e^{2x} - 1 = 2e^x \Rightarrow e^{2x} - 2e^x - 1 = 0$

$y^2 - 2y - 1 = 0$ Let $y = e^x$

$y = \frac{2 \pm \sqrt{4+4}}{2} = \frac{2 \mp 2\sqrt{2}}{2} \Rightarrow e^x = 1 \pm \sqrt{2}$

$e^x = 1 + \sqrt{2}$

$\ln e^x = \ln(1+\sqrt{2})$

$x = \ln(1+\sqrt{2})$ ✓✓

$e^x = 1 - \sqrt{2}$ impossible

Ex: if $f(x) = \sqrt{3 - e^{2x}}$, find $f^{-1}(x)$?

Sol:- $y = \sqrt{3 - e^{2x}}$

$y^2 = 3 - e^{2x} \rightarrow e^{2x} = 3 - y^2 \Rightarrow \frac{2x}{2} = \frac{\ln(3-y^2)}{2} = x = \frac{\ln(3-y^2)}{2}$

$f^{-1}(x) = \frac{\ln(3-x^2)}{2}$

$= \ln\sqrt{3-x^2}$

Ex: if $f(x) = \ln(2+\ln x)$, find $f^{-1}(x)$?

Sol: $y = \ln(2+\ln x)$

$e^y = 2+\ln x$

$e^y - 2 = \ln x$

$e^{e^y-2} = x$

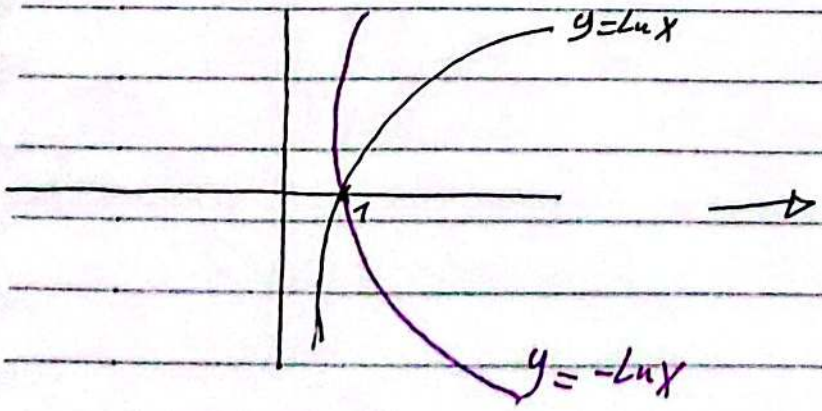
$f^{-1}(x) = e^{e^x-2}$

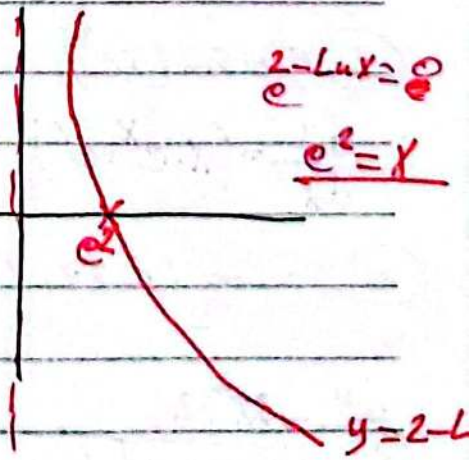
Q Solve $xe^{-x} + 3e^{-x} = 0$!-

$e^{-x}(x+3) = 0$

$x+3=0$ $\boxed{x=-3}$ $e^{-x} = 0$

Ex: Stretch $y = 2-\ln x$ and find the domain Rang?





D: $(0,\infty)$

R: R

Amazing

6.2 + 6.3 + 6.4 + 6.6

Rules:

① $\frac{d}{dx}\left(b^{f(x)}\right) = f'(x) \cdot b^{f(x)} \cdot \ln b$

② $\frac{d}{dx}\left(e^{f(x)}\right) = f'(x) \cdot e^{f(x)}$

③ $\frac{d}{dx}\left(\log_b f(x)\right) = \frac{f'(x)}{f(x) \cdot \ln b}$

④ $\frac{d}{dx}\left(\ln f(x)\right) = \frac{f'(x)}{f(x)}$

Ex: Find $\frac{dy}{dx}$ of the following:

① $y = 5^{x^2} + e^{\sqrt{x+1}}$

Sol: $\frac{dy}{dx} = 2x \cdot 5^{x^2} \cdot \ln 5 + \frac{1}{2\sqrt{x+1}} \cdot e^{\sqrt{x+1}}$

② $y = \sin(xe^x)$

$\frac{dy}{dx} = (x \cdot e^x + e^x) \cdot \cos(xe^x)$

③ $y = \log_2(x^3 + 8x + 1)$

$\frac{dy}{dx} = \frac{3x^2 + 8}{(x^3 + 8x + 1) \cdot \ln 2}$

④ $y=(\log \frac{1}{x})^3$

$$\frac{dx}{dy}=3(\log \frac{1}{x})^2 \cdot \frac{\frac{1}{x^2}}{\frac{1}{x}\cdot Ln\,10}=\frac{-3}{x\cdot Ln\,10}\cdot(\log \frac{1}{x})^2$$

⑤ $y=\cos x \cdot Ln(\sin x)$

$$\frac{dy}{dx}=\cos x\cdot\frac{\cos x}{\sin x}-Ln(\sin x)(\sin x)$$

⑥ $y^2=Ln(Ln(e^x+1))$

$$2yy'=\frac{\frac{e^x}{e^x+1}}{Ln(e^x+1)} \Rightarrow \frac{dy}{dx}=\frac{e^x}{2y(e^x+1)Ln(e^x+1)}$$

⑦ $y=x^{\sin x}$

Sol:- $Lny=Ln\,x^{\sin x} \longrightarrow Lny=\sin x\,Ln\,x$

$$\frac{1}{y}\cdot y'=\sin x\cdot\frac{1}{x}+Ln\,x\cos x$$

$$\frac{dy}{dx}=y\left(\frac{\sin x}{x}+\cos x\,Ln\,x\right)$$

$$=x^{\sin x}$$

⑧ $y=(x^2+1)^{Lnx}$

$$Lny=Ln(x^2+1)\cdot Ln\,x$$

$$\frac{y'}{y}=Ln(x^2+1)\cdot\frac{1}{x}+Ln\,x\cdot\frac{2x}{x^2+1}$$

$$\frac{dy}{dx}=\left(Ln(x^2+1)\cdot\frac{1}{x}+Ln\,x\cdot\frac{2x}{x^2+1}\right)\cdot y$$

Ex:- if $F(x)=\int_1^x Ln(t^2+1)\,dt$, find $F^{**}(1)$?

Sol:- $F'(x)=\frac{d}{dx}\int_1^x Ln(t^2+1)dt = Ln(x^2+1)+Ln2$

$F''(x)=\frac{2x}{x^2+1} \longrightarrow F''(1)=1$

Q if $F(x)=\int_x^{e^x}\sqrt{t+1}\,dt$, find $F'(x)$

§ $\sqrt{e^x+1}-\sqrt{x+1}$

Rules①

② $\int K^{ax+b}\,dx=\frac{1}{a}\frac{K^{ax+b}}{Ln\,K}$

② $\int e^{ax+b}\,dx=\frac{1}{a}e^{ax+b}+C$

③ $\int\frac{1}{ax+b}\,dx=\frac{1}{a}Ln|ax+b|+C$

④ $\int\frac{f'(x)}{f(x)}\,dx=Ln|f(x)|+C$

Ama

Ex: Evaluate the following

① $\int 2^{3x-1}\,dx = \frac{1}{3}\cdot\frac{2^{3x-1}}{\ln 2} + C$

② $\int \sin 2x \cdot e^{\sin^2 x}\,dx$ = Soli- Let $y = \sin^2 x$

$\frac{dy}{dx} = 2\sin x \cdot \cos x$

$\frac{dx}{dy} = \frac{1}{\sin 2x}$

$dx = \frac{dy}{\sin 2x}$

$= \int \sin 2x \cdot e^{y} \cdot \frac{dy}{\sin 2x} = e^{y} + C$

$= e^{\sin^2 x} + C$

② $\int 5^x \cdot 2^{2x}\,dx$

$\int 5^x (2^2)^x\,dx \Rightarrow \int (5\cdot 4)^x\,dx \Rightarrow \int 20^x\,dx = \frac{20^x}{\ln 20} + C$

④ $\int \frac{5}{2x-3}\,dx = 5\int \frac{1}{2x-3}\,dx \Rightarrow 5\cdot\frac{1}{2}\cdot\ln|2x-3| + C$

⑤ $\int \frac{2x+1}{x^2+x+5}\,dx = \ln|x^2+x+5| + C$

⑥ $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\cos x}\,dx \Rightarrow -\int_0^{\frac{\pi}{2}} \frac{-\sin x}{1+\cos x}\,dx \Rightarrow = -\ln|1+\cos x|\Big|_0^{\frac{\pi}{2}}$

$= -\ln(1) + \ln 2$

$= \ln 2$

⑦ $\int \frac{(\ln x)^2}{x}\, dx$

Sol: $y = \ln x \rightarrow \frac{dy}{dx} = \frac{1}{x}$

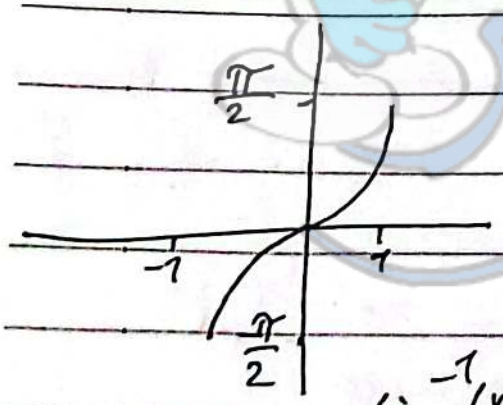
ومنه

$$dx = x\,dy$$

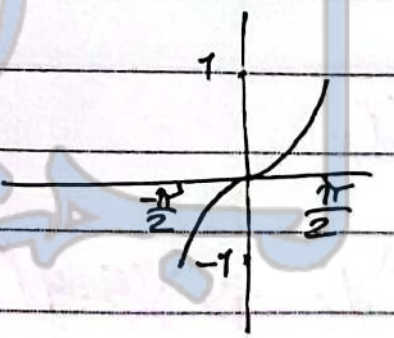
$$= \int \frac{y^2}{x} \cdot x\, dy = \frac{y^3}{3} + C \Rightarrow \frac{\ln^3}{3} + C$$

6,7 Inverse Trigonometric Functions:-

② if $f(x) = \sin x, \frac{-\pi}{2} \leq x \leq \frac{\pi}{2}$ then $f^{-1}(x) = \sin^{-1}(x)\ -1 \leq x \leq 1$



$\sin^{-1}(x)$



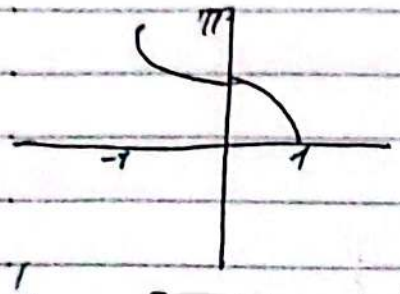
$\sin(x)$

Dan and Rany

بكلمو بحكي بالإنجليزي

② if $f(x) = \cos x$, $0 \le x \le \pi$ then

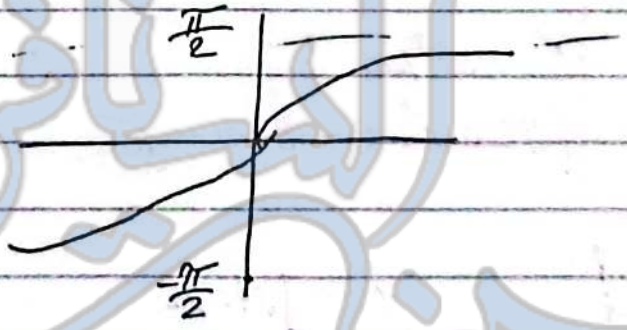
$f^{-1}(x) = \cos^{-1} x$, $-1 \le x \le 1$



Dom $= [-1, 1]$ and Rang $= [0, \pi]$

③ if $f(x) = \tan x$, $\frac{-\pi}{2} < x < \frac{\pi}{2}$ then

$f^{-1}(x) = \tan^{-1} x$



Dom: R

Rang $= \left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

④ if $f(x) = \cot x$, $0 \le x \le \pi$, then

$f^{-1}(x) = \cot^{-1}(x)$

⑤ if $f(x) = \sec$, $0 \le x \le \pi$ and $x \neq \frac{\pi}{2}$, then

$f^{-1}(x) = \sec^{-1} x$

⑥ if $f(x) = \csc x$, $\frac{-\pi}{2} < 0 < \frac{\pi}{2}$ and $x \neq 0$ then

$f^{-1}(x) = \csc^{-1} x$

Amazing

Rules:- ① $\sin^{-1}(-x) = -\sin^{-1}(x)$

② $\cos^{-1}(-x) = \pi - \cos^{-1}x$

③ $\tan^{-1}(-x) = -\tan^{-1}x$

④ $\sin^{-1}(x) + \cos^{-1}x = \frac{\pi}{2}$

Ex: Evalvate the following

① $\sin^{-1}\left(\frac{1}{2}\right)$

Sol:- let $y = \sin^{-1}\left(\frac{1}{2}\right)$

$\sin y = \sin\left(\sin^{-1}\frac{1}{2}\right)$

$\sin y = \frac{1}{2}$

$y = \frac{\pi}{6} \Rightarrow \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$

② $\sin^{-1}(-1)$

$= -\sin^{-1}(1) = -\frac{\pi}{2}$

② $\tan^{-1}(1) = \frac{\pi}{4}$

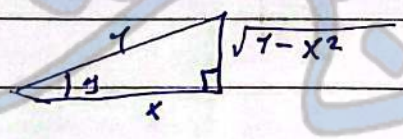
④ $\cos^{-1}\left(\frac{-1}{2}\right) = \frac{\pi}{3}$

⑤ $\cos^{-1}(-1)$

$\pi - \cos^{-1}(-1) = \pi - 0 = \pi$

⑥ $\tan(\cos^{-1}x) = ?$

Sol: let $y = \cos^{-1}x$

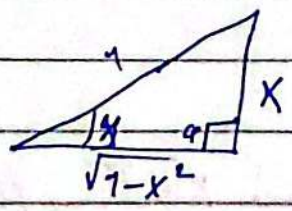


$\cos y = x = \frac{x}{1}$

$\tan(\cos^{-1}x) = \tan y = \frac{\sqrt{1-x^2}}{x}$

⑦ $\sin(2\sin^{-1}x)$

Sol: let $y = \sin^{-1}x$



$\sin y = x = \frac{x}{1}$

$\sin(2\sin^{-1}x) = \sin 2y = 2\sin y \cdot \cos y = 2x$

$= 2x\frac{\sqrt{1-x^2}}{1}$

An

Theorem:-

① $\frac{d}{dx}\sin^{-1}(f(x)) = \frac{f'(x)}{\sqrt{1-(f(x))^2}}$

② $\frac{d}{dx}\cos^{-1}(f(x)) = \frac{-f'(x)}{\sqrt{1-(f(x))^2}}$

③ $\frac{d}{dx}\tan^{-1}(f(x)) = \frac{f'(x)}{1+f(x)^2}$

④ $\frac{d}{dx}\cot^{-1}(f(x)) = \frac{-f'(x)}{1+f(x)^2}$

⑤ $\frac{d}{dx}\sec^{-1}(f(x)) = \frac{f'(x)}{|f(x)|\cdot\sqrt{f(x)^2-1}}$

⑥ $\frac{d}{dx}\csc^{-1}(f(x)) = \frac{-f'(x)}{|f(x)|\cdot\sqrt{f(x)^2-1}}$

Ex: Find $\frac{dy}{dx}$ of:-

① $y = \cos^{-1}(x+1) + \tan^{-1}(x^3)$

$\frac{dy}{dx} = \frac{-1}{\sqrt{1-(x+1)^2}} + \frac{3x^2}{1+(x^3)^2}$

② $y = 2^{\sin^{-1}(\frac{1}{x})} + \cot^{-1}(e^x)$

$\frac{dy}{dx} = \frac{-\frac{1}{x^2}}{\sqrt{1-(\frac{1}{x})^2}} \cdot 2^{\sin^{-1}(\frac{1}{x})} \cdot \ln 2 + \frac{-e^x}{1+(e^x)^2}$

② $y = \ln(\sec^{-1}(2x)) = \frac{\frac{2}{|2x|\cdot\sqrt{(2x)^2-1}}}{\sec^{-1}(2x)}$

④ $y = \cos^{-1}(\cos x) = \frac{\sin(x)}{\sqrt{1-\cos^2 x}} = \frac{\sin x}{|\sin x|} = \frac{\sin x}{\sin x} = 1$

, Lפוֹל:

$y = x$

$\frac{dy}{dx} = 1$

Amazing

⑤ $y=\sin(\tan^{-1}(\sqrt{x}))$

$$\frac{dy}{dx}=\frac{\frac{1}{2\sqrt{x}}}{1+\sqrt{x}^2}\cdot\cos(\tan^{-1}(\sqrt{x}))$$

$$=\frac{1}{4}$$

⑥ $y=(x^2+1)\cdot\csc^{-1}(x)$

$$-(x^2+1)\frac{1}{x\sqrt{x^2-1}}+\csc^{-1}(x)\cdot 2x$$

⑦ $y=\left(\sin^{-1}\left(\frac{x+1}{x-1}\right)\right)^3 \Rightarrow 3\left(\sin^{-1}\left(\frac{x+1}{x-1}\right)\right)^2\cdot\frac{\frac{(x-1)-(x+1)}{(x-1)^2}}{\sqrt{1-\left(\frac{x+1}{x-1}\right)^2}}$

Rules:-

① $\int\frac{1}{\sqrt{a^2-x^2}}dx=\sin^{-1}\left(\frac{x}{a}\right)+c$ ضيف سالب $\longrightarrow$ $\cos^{-1}$

② $\int\frac{1}{a^2+x^2}dx=\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+c$ $\xrightarrow{m}$ $\cot^{-1}$

③ $\int\frac{1}{x\sqrt{x^2-a^2}}dx=\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right)+c$ $\xrightarrow{m}$ $\csc^{-1}$

Ex:- Evaluate the following:-

① $\int \frac{dx}{2+x^2} = \int \frac{dx}{(\sqrt{2})^2+x^2} = \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right)+C$ ✓

② $\int \frac{\cos x}{\sqrt{4-\sin^2 x}}dx =$ Let $\sin x = y$ $\frac{dy}{dx} = \cos x$ ✓

$\int \frac{\cos x}{\sqrt{4-y^2}}\frac{dy}{\cos x} = \int \frac{dy}{\sqrt{2^2-y^2}} = \sin^{-1}\left(\frac{y}{2}\right)+C \Rightarrow \sin^{-1}\left(\frac{\sin x}{2}\right)+C$

③ $\int \frac{x}{x^4+1}dx = \int \frac{x}{(x^2)^2+1}dx$ Let $x^2=y$ $\frac{dy}{dx}=2x$ ✓

$\int \frac{x}{y^2+1}\frac{dy}{2x} = \frac{1}{2}\tan^{-1}\left(\frac{y}{1}\right)+C = \frac{1}{2}\tan^{-1}(x^2)+C$

④ $\int \frac{dx}{x\sqrt{4x^2-9}} = \frac{1}{2}\int \frac{dx}{x\sqrt{x^2-\left(\frac{3}{2}\right)^2}} = \frac{1}{2}\times\frac{2}{3}\times\sec^{-1}\left(\frac{2x}{3}\right)+C$ ✓

6.8 Hyperbolic Functions

$e^x = \frac{e^x+e^{-x}}{2} + \frac{e^x-e^{-x}}{2}$

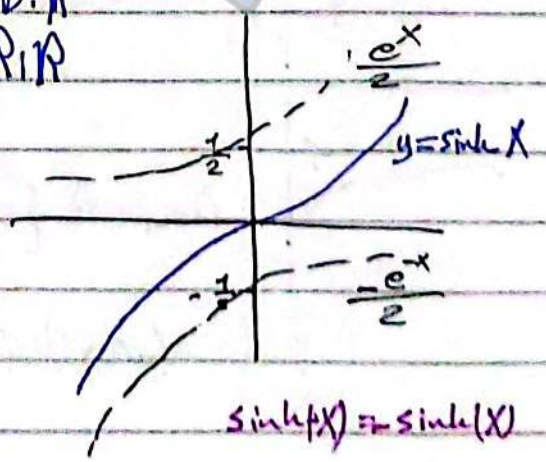
even odd

def:-

① $\sinh x = \frac{e^x-e^{-x}}{2}$

D: $\mathbb{R}$

R: $\mathbb{R}$



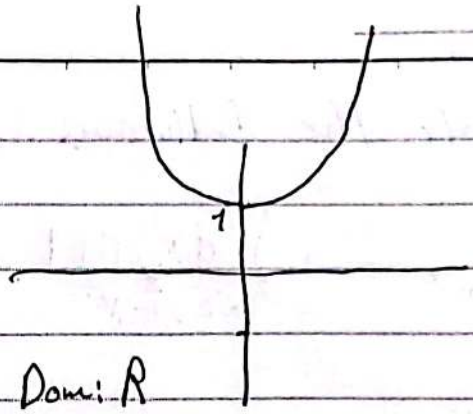
$\sinh(-x) = -\sinh(x)$

$\cosh + \sinh = e^x$

$\cosh - \sinh = e^{-x}$

Amazing

② $\cosh x = \frac{e^x + e^{-x}}{2}$

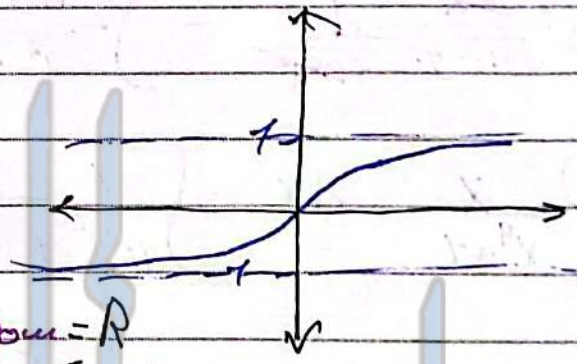


$\cosh(-x) = \cosh x$

Dom: $\mathbb{R}$

$\mathbb{R} = [1, \infty)$

③ $\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

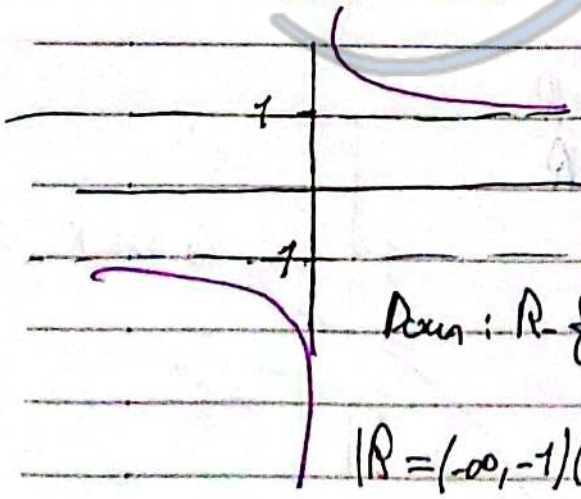


$\tanh(-x) = -\tanh(x)$

Dom = $\mathbb{R}$

$\mathbb{R} = (-1, 1)$

④ $\coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

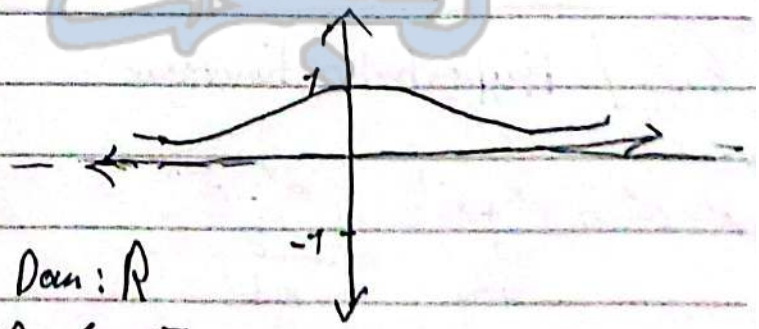


Dom : $\mathbb{R} - \{0\}$

$\mathbb{R} = (-\infty, -1) \cup (1, \infty)$

$\coth(-x) = -\coth(x)$

⑤ $\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$



Dom : $\mathbb{R}$

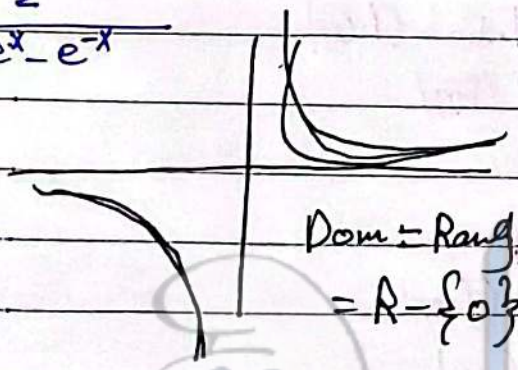
$\mathbb{R} = (0, 1]$

$\operatorname{sech}(-x) = \operatorname{sech} x$

Ami

④ ~~coth~~ $\coth x = \frac{1}{\sinh x}$

$= \frac{2}{e^x - e^{-x}}$



Dom = Rang

$= R - \{0\}$

$\coth(-x) = -\coth(x)$

Identities:

① $\cosh^2 x - \sinh^2 x = 1$

② $1 - \tanh^2 x = \operatorname{sech}^2 x$

③ $\coth^2 x - 1 = \operatorname{csch}^2 x$

④ $\sinh(2x) = 2\sinh x \cosh x$

⑤ $\cosh(2x) = \cosh^2 x + \sinh^2 x$

$= 1 + 2\sinh^2 x$

$= 2\cosh^2 x - 1$

Ex: if $\sinh(a) = \sqrt{8}$, find ① $\cosh(a)$ ② $\cosh(4a)$

① $\cosh^2 a - \sinh^2 a = 1$

$\cosh^2 a - (\sqrt{8})^2 = 1$

$\sqrt{\cosh^2 a} = \sqrt{9}$

$\cosh a = 3$ ~~-3~~ ~~X~~

② $\cosh 4a = \cosh^2(2a) + \sinh^2(2a)$

$\cosh(2a) = \sinh^2(a) + \cosh^2(a)$

$8 + 9$

$\cosh(2a) = 17$

$\sinh(2a) = 2\sinh a \cosh a$

$2 \times 3 \times \sqrt{8}$

$\cosh(4a) = (17)^2 + (6\sqrt{8})^2$

Rules: ① $\frac{d}{dx}\sinh(f(x)) = f'(x)\cdot\cosh(f(x))$

② $\frac{d}{dx}\cosh(f(x)) = f'(x)\cdot\sinh(f(x))$

③ $\frac{d}{dx}\tanh(f(x)) = f'(x)\cdot\operatorname{sech}^2(f(x))$

④ $\frac{d}{dx}\coth(f(x)) = -f'(x)\operatorname{csch}^2(f(x))$

⑤ $\frac{d}{dx}\operatorname{sech}(f(x)) = -f'(x)\operatorname{sech}(f(x))\tanh(f(x))$

⑥ $\frac{d}{dx}\operatorname{csch}(f(x)) = -f'(x)\operatorname{csch}(f(x))\coth(f(x))$

Ex: Find $\frac{dy}{dx}$ of the following:

① $y = \cosh(x^3-8) + \sinh(e^{2x+1})$

$\frac{dy}{dx} = 3x^2\sinh(x^3-8) + 2e^{2x+1}\cdot\cosh(e^{2x+1})$

② $y = \frac{\operatorname{sech}(\sqrt{x})}{x} \Rightarrow \frac{dy}{dx} = \frac{x\cdot\frac{-1}{2\sqrt{x}}\operatorname{sech}(\sqrt{x})\tanh\sqrt{x} - \operatorname{sech}(\sqrt{x})}{x^2}$

③ $y^2 = \ln(\tanh(3x))$

$\frac{dy}{dx}\cdot 2y = \frac{3\cdot\operatorname{sech}^2(3x)}{\tanh(3x)} = \frac{3\cdot\operatorname{sech}^2(3x)}{\tanh(3x)\cdot 2y}$

④ $y = \sin^{-1}(\coth x)$

$\frac{dy}{dx} = \frac{-\operatorname{csch}^2 x}{\sqrt{1-\coth^2 x}}$

⑤ $y = \sinh^3\left(\frac{1}{x}\right) = 3\sinh^2\left(\frac{1}{x}\right)\cdot\frac{-1}{x^2}\cdot\cosh\frac{1}{x}$

⑥ $y = (\sinh x)^{\cosh x}$

$\ln y = \ln(\sinh(x))^{\cosh(x)}$

$\ln y = \cosh(x)\cdot\ln(\sinh(x))$

$\frac{1}{y}\frac{dy}{dx} = \cosh(x)\cdot\frac{\cosh x}{\sinh(x)} + \sinh x\cdot\ln(\sinh x)$

Ex: $\cosh(0) = \frac{e^0+e^0}{2} = 1$

$\sinh(0) = \frac{e^0-e^0}{2} = 0$

$\tanh(2) = \frac{e^2-e^{-2}}{e^2+e^{-2}}$

$= \left(\frac{e^2-e^{-2}}{e^2+e^{-2}}\right)\frac{e^2}{e^2} = \frac{e^4-1}{e^4+1}$

Amazing

Rules: ① $\int \sinh(ax+b)\,dx = \frac{1}{a}\cosh(ax+b)+C$

② $\int \cosh(ax+b)\,dx = \frac{1}{a}\sinh(ax+b)+C$

③ $\int \operatorname{sech}^2(ax+b)\,dx = \frac{1}{a}\tanh(ax+b)+C$

④ $\int \operatorname{csch}^2(ax+b)\,dx = \frac{-1}{a}\coth(ax+b)+C$

⑤ $\int \operatorname{sech}(ax+b)\cdot\tanh(ax+b)\,dx = \frac{-1}{a}\operatorname{sech}(ax+b)+C$

⑥ $\int \operatorname{csch}(ax+b)\coth(ax+b)\,dx = \frac{-1}{a}\operatorname{csch}(ax+b)+C$

Ex1- ① $\int \coth^2(2x-7)\,dx$

$= \int \operatorname{csch}^2(2x-7)+1\,dx$

$= \frac{-1}{2}\coth(2x-7)+x+C$

② $\int \tanh(3x)\,dx = \frac{1}{3}\int \frac{3\sinh(3x)}{\cosh(3x)}\,dx$

$= \frac{1}{3}\ln|\cosh(3x)|+C$

③ $\int \sinh^4 x\cdot\cosh x\,dx$ | let $y=\sinh x$, $dy=\cosh x\,dx$

$\int y^4\cdot\cosh x\,\frac{dy}{\cosh x}$

$\int y^4\,dy = \frac{y^5}{5}+C$

$= \frac{1}{5}\sinh^5 x+C$

④ $\int_{\ln 2}^{\ln 3} \tanh x\cdot\operatorname{sech}^3 x\,dx$ | $y=\operatorname{sech} x$, $dy=-\operatorname{sech} x\tanh x$

$\int_{\ln 2}^{\ln 3} (\tanh x\cdot\operatorname{sech})\,y^2\cdot\frac{dy}{-\operatorname{sech} x\tanh x}$

$y = \frac{2}{e^{\ln 2}+e^{-\ln 2}} = \frac{2}{\frac{5}{2}} = \frac{4}{5}$ → $y = \frac{2}{3+\frac{1}{3}} = \frac{6}{10}$

$= \int_{\frac{4}{5}}^{\frac{6}{10}} -y^2\,dy = \frac{-y^3}{3}\Big|_{\frac{4}{5}}^{\frac{6}{10}}$

⑤ $\int \operatorname{sech} x\,dx = \int \frac{2}{e^x+e^{-x}} \cdot \frac{e^x}{e^x}\,dx$

$= \int \frac{2e^x}{e^{2x}+1}\,dx \Rightarrow$ Let $y = e^x \rightarrow dx = \frac{dy}{e^x}$

$\int \frac{2}{y^2+1} \cdot \frac{dy}{y} = 2 \cdot \int \frac{1}{y^2+1^2}\,dy = 2 \cdot 1 \cdot \tan^{-1}\left(\frac{y}{1}\right) + C = 2\tan^{-1}(e^x) + C$

6.5 L'Hopital's Rule

Theorem (L.R): if $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ are exist and

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \begin{cases} \frac{0}{0}, \frac{\infty}{\infty}, 0\cdot\infty, \infty-\infty \\ 0^0, \infty^0, 1^\infty \end{cases}$$

Then $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$

Note: This statement is true when $x \to a^+$, $x \to a^-$, $x \to \infty$, $x \to -\infty$

Ex: Find the following:

① $\lim_{x\to 1} \frac{x^2-2x+1}{x^3-1} \Rightarrow \lim_{x\to 1} \frac{2x-2}{3x^2} = \frac{0}{3} = 0$

② $\lim_{x\to\infty} \frac{x^3+x+7}{x^4+x^2} \overset{\lim_{x\to\infty}}{\Rightarrow} \frac{3x^2+1}{4x^3+2x} \Rightarrow \lim_{x\to\infty} \frac{6x}{12x^2+2} \Rightarrow \lim_{x\to\infty} \frac{6}{24x} = \frac{6}{\infty} = 0$

③ $\lim_{x\to\pi} \frac{x\cdot\sin x}{x-\pi} \overset{\lim_{x\to\pi}}{\Rightarrow} \frac{x\cdot\cos x + \sin x}{1} = \frac{-\pi\cdot 1 + 0}{1} = \pi$

$\frac{\infty}{\infty} = 0$

Amazing

④

$\lim_{x\to\infty} \frac{\ln x}{e^x} \Rightarrow \lim_{x\to\infty} \frac{\frac{1}{x}}{e^x} = 0$

⑤ $\lim_{x\to\infty} \frac{e^x - 1}{e^x + x} \overset{\lim}{\Rightarrow} \frac{e^x}{e^x+1} = \lim_{x\to\infty} \frac{e^x}{e^x} = 1$

$f(x) = e^{\ln f(x)}$

$\lim_{x\to a} f(x) = \lim_{x\to a} e^{\frac{\ln f(x)}{f(g(x))}}$

① $\lim_{x\to a} f(g(x)) = f(\lim_{x\to a} g(x))$

$\lim_{x\to a} f(x) = \lim_{y\to a} e^{\ln f(x)}$

$= e^{\lim_{x\to a} \ln f(x)}$

⑥ $\lim_{x\to 0} (x)^x \Rightarrow \lim_{x\to 0} e^{x\ln(x)} = \lim_{x\to 0} e^{\frac{\ln(x)}{\frac{1}{x}}}$

$= e^{\lim_{x\to 0} \frac{\frac{1}{x}}{\frac{-1}{x^2}}} = e^{\lim_{x\to 0} -x} = e^0 = 1$

⑦ $\lim_{x\to\infty} (2+x)^{\frac{1}{x}} = \infty^0$

$= e^{\lim_{x\to\infty} \frac{1}{x}\ln(2+x)} = e^{\lim_{x\to\infty} \frac{\frac{1}{2+x}}{1}}$

$= e^{\frac{1}{2+\infty}} = e^0 = 1$

⑧ $\lim_{x\to 0^+} (\cos x)^{\frac{1}{x}} \Rightarrow e^{\lim_{x\to 0^+} \frac{1}{x}\ln\cos x}$

$= e^{\lim_{x\to 0^+} \frac{\frac{-\sin x}{\cos x}}{1}} = e^0 = 1$

$\tan 0 = 0$

⑨ $\lim_{x\to 0} (1+x)^{\frac{1}{x}}$

$e^{\lim_{x\to 0} \ln(1+x)\cdot\frac{1}{x}} = e^{\frac{\frac{1}{1+x}}{1}} = e^1$ ✓✓

Rule $\lim_{x\to 0} (1+x)^{\frac{k}{x}} = e^k$

7

CH 7:

7.2 Integration by Parts.

$\int f(x).g(x)\,dx = f(x).G(x) - \int f'(x)\,G(x)\,dx$ ..... ①

where $G(x) = \int g(x)\,dx$. this formula is called integration by parts.

We usually rewrite ① by letting:

$u = f(x) \rightarrow \frac{du}{dx} = f'(x) \rightarrow du = f'(x)\,dx$

$v = G(x) \rightarrow \frac{dv}{dx} = G'(x) = g(x) \rightarrow dv = g(x)\,dx$

$$\int u.dv = u.v - \int v.du \quad \text{---} ②$$

Ex: Evaluate the following:-

① $\int x.\sinh(2x)\,dx$

$\frac{du}{dx} = 1 \quad du = dx$

$dv = \sinh(2x)\,dx \Rightarrow \int \frac{dv}{dx} = \int \sinh 2x\,dx$

$v = \frac{\cosh(2x)}{2}$

$\int x.\sinh(2x)\,dx = \frac{x}{2}\cosh(2x) - \int \frac{1}{2}\cosh(2x).dx$

$= \frac{x}{2}.\cosh(2x) - \frac{1}{2\times 2}\sinh(2x) + C$

OR

	جزئی	تکلی
+	x	$\sinh(2x)$
-	1	$\frac{1}{2}\cosh(2x)$
+	0	$\frac{1}{4}\sinh(2x)$

$\int x.\sinh(2x).dx$

$= \frac{x}{2}\cosh(2x) - \frac{1}{4}\sinh(2x) + C$

Amazing

② $\int \underbrace{(x^3+1)}_{u} \cdot \underbrace{e^{2x+1}\,dx}_{dv}$

$u = x^3+1 \longrightarrow du = 3x^2\,dx$

$e^{2x+1} \longrightarrow \frac{e^{2x+1}}{2}$

او بالطريقة الأسهل :

	u	du
+	x^3+1	e^{2x+1}
−	$3x^2$	$\frac{e^{2x+1}}{2}$
+	$6x$	$\frac{e^{2x+1}}{4}$
−	6	$\frac{e^{2x+1}}{8}$
+	0	$\frac{e^{2x+1}}{16}$

$(x^3+1)\frac{e^{2x+1}}{2} - \frac{3x^2}{4}e^{2x+1} + \frac{6x}{8}e^{2x+1} - \frac{6}{16}e^{2x+1} + C$

③ $\int \underbrace{\cos^{-1}(2x)}_{u} \cdot \underbrace{dx}_{dv}$

$u = \cos^{-1}(2x) \Rightarrow du = \frac{-2}{\sqrt{1-4x^2}}$

$dv = dx \Rightarrow v = x$

$= x\cdot\cos^{-1}(2x) + \int \frac{2x}{\sqrt{1-4x^2}}\,dx$ ⟵ Let $y = 1-4x^2 \Rightarrow dx = \frac{dy}{-8x}$

$= x\cdot\cos^{-1}(2x) + \int \frac{2x}{\sqrt{y}}\cdot\frac{dy}{-8x} = x\cdot\cos^{-1}(2x) + \int \frac{1}{4}y^{-\frac{1}{2}}\,dx$

④ $\int \sin(\ln x)\,dx$ let $y = \ln x$, $dy = \frac{1}{x}dx$

$\int \sin y\,dy \cdot x$ $\boxed{e^y = x}$

$\int \sin y \cdot e^y\,dy$

$u = e^y \longrightarrow du = e^y\,dy$

$dv = \int \sin y\,dy \longrightarrow v = -\cos y$

$= -e^y\cdot\cos y + \int e^y \cos y\,dy$

$u = e^y \longrightarrow du = e^y\,dy$

$dv = \int \cos y \Rightarrow v = \sin y$

$= -e^y\cdot\cos y + e^y\sin y - \int e^y \sin y\,dy$

$2\int \sin(\ln x)\,dx = -e^y\cos y + e^y\sin y$

$\int \sin(\ln x)\,dx = \frac{-e^y\cos y + e^y\sin y}{2}$

$= \frac{-x\cos(\ln x) + x\sin(\ln x)}{2}$

Amazing

5) $\int \frac{xe^x}{(x+1)^2}\,dx$

sol: $u = xe^x \longrightarrow du = xe^x + e^x$

$\longrightarrow du = (x+1)e^x\,dx$

$dv = \frac{1}{(x+1)^2}\,dx \Rightarrow \frac{dv}{dx} = \frac{1}{(x+1)^2}$

$v = \int \frac{1}{(x+1)^2}\,dx = \int (x+1)^{-2}\,dx$

$= \frac{-1}{x+1}$

$\therefore \int \frac{xe^x}{(x+1)^2}\,dx = \frac{-xe^x}{x+1} + \int \frac{1}{x+1}\cdot(x+1)e^x\,dx$

$= \frac{-xe^x}{x+1} + e^x + C$

6) $\int \sqrt{x+1}\,(x^2+2x+5)\,dx$

7) $\int \cos(\sqrt{x})\,dx$

8) $\int \ln x\,dx$

9) $\int \frac{\ln x}{\sqrt{x}}$

7.3

Integrating Trigonometric Functions:-

Rule:

① $\int \sin^n x\,dx = \frac{-1}{n}\sin^{n-1}x\cdot\cos x + \frac{n-1}{n}\int \sin^{n-2}x\,dx$

② $\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1}x\cdot\sin x + \frac{n-1}{n}\int \cos^{n-2}x\,dx$

③ $\int \tan^n x\,dx = \frac{\tan^{n-1}x}{n-1} - \int \tan^{n-2}x\,dx$

④ $\int \sec^n x\,dx = \frac{\sec^{n-2}x\cdot\tan x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2}x\,dx$

ex:-

① $\int \cos^5 x\,dx = \frac{1}{5}\cos^4\cdot\sin x + \frac{4}{5}\int \cos^3 x\,dx$

$= \frac{1}{5}\cos^4 x\cdot\sin x + \frac{4}{5}\left(\frac{1}{3}\cos^2 x\cdot\sin x + \frac{2}{3}\int \cos x\,dx\right)$

$= \frac{1}{5}\cos^4 x\cdot\sin x + \frac{4}{15}\cos^2 x\cdot\sin x + \frac{8}{15}\sin x + C$

② $\int \sec^4 x\,dx = \frac{\sec^2 x\cdot\tan x}{3} + \frac{2}{3}\int \sec^2 x\,dx$

$= \frac{1}{3}\sec^2\cdot\tan x + \frac{2}{3}\tan x + C$

Amazing

③ $\int \tan^3(2x+1)\,dx \Rightarrow$ let $y = 2x+1$

$dy = 2\,dx$

$= \frac{1}{2}\int \tan^3 y\,dy = \frac{1}{2}\left(\frac{\tan^2 y}{2} - \int \tan y\,dy\right)$

$= \frac{1}{4}\tan^2 y + \frac{1}{2}\ln|\cos y| + C =$ [2a, الحل]

④: $\int x \cdot \sin^3 x\,dx$

+	x	$\sin^3 x$
−	1	$-\frac{1}{3}\sin^2 x\cos x + \frac{2}{3}\int \sin x\,dx$
	0	$-\frac{1}{3}\sin^3 x - \frac{2}{3}\cos x$

$\int -\frac{1}{3}\sin^2 x \cdot \cos x$ | $\sin x = y$, $\cos x = dy$

$-\int \frac{1}{3} y^2 \cdot \cos x \cdot \frac{dy}{\cos x}$

$= -\frac{1}{9}\sin^3 x$

$= x\left(-\frac{1}{3}\sin^2 x\cos x - \frac{2}{3}\cos x\right) + \frac{1}{9}\sin^3 x + \frac{2}{3}\sin x + C$

⑤ $\int \sin^m x \cdot \cos^n x\,dx = ?$

① if $n = 2k+1$ (odd), save one cosine function and use $\cos^2 x = 1 - \sin^2 x$ to express the remaining factors in terms of sine:

$\int \sin^m x \cdot \cos^{2k+1} x\,dx = \int \sin^m x \cdot \cos^{2k} x \cdot \cos x\,dx$

$= \int \sin^m x \cdot (1-\sin^2 x)^k \cdot \cos x\,dx$

Then substitute $y = \sin x$

② if $m = 2k+1$ (odd), save "sine" and $\sin^2 x = 1 - \cos^2 x$ to express the remaining factors in terms of "cosine".

$\int \sin^{2k+1} x \cdot \cos^n x\,dx = \int \sin x \cdot \underbrace{\sin^{2k} x}_{(\sin^2 x)^k} \cdot \cos^n x\,dx$

$= \int \sin x \cdot (1-\cos^2 x)^k \cdot \cos^n x\,dx$

let $y = \cos x$

③ if the powers of both sine and cosine are odd, either ① or ② can be used.

④ if the powers of both sin/cos are even, use:-

$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$

$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$

Amazing

① $\int \sin^6 x \cdot \cos^3 x\, dx$ (3 → odd)

$= \int \sin^6 x \cdot \cos^2 x \cdot \cos x\, dx$

$= \int \sin^6 x \cdot (1-\sin^2 x) \cdot \cos x\, dx$ | let $y = \sin x$, $dy = \cos x\, dx$

$= \int y^6 \cdot (1-y^2) \cdot \cos x \frac{dy}{\cos x}$

$= \int y^6 - y^8\, dy = \frac{y^7}{7} - \frac{y^9}{9} + c$

$= \frac{\sin^7 x}{7} - \frac{\sin^9 x}{9} + c$

② $\int \sin^5 x \cdot \cos^2 x\, dx$

$\int \sin x \cdot \sin^4 x \cdot \cos^2 x\, dx$

$= \int \sin x \cdot (1-\cos^2 x)^2 \cdot \cos^2 x\, dx$ | let $y = \cos x$, $dy = -\sin x\, dx$

$= \int \sin x \cdot (1-y^2)^2 \cdot y^2 \frac{dy}{-\sin x}$

$= -\int (1-2y^2+y^4) \cdot y^2\, dy = \int y^2 - 2y^4 + y^6\, dy$

$= -\frac{y^3}{3} + \frac{2}{5}y^5 - \frac{y^7}{7} + c$

$= -\frac{\cos^3 x}{3} + \frac{2}{5}\cos^5 x - \frac{\cos^7 x}{7} + c$

③ $\int \sin^3 x \cdot \cos^5 x\, dx$

$\int \sin x \cdot \sin^2 x \cdot \cos^5 x\, dx$ | let $y = \cos x$

$\int \sin x \cdot (1-\cos^2 x) \cdot \cos^5 x\, dx$ | $dy = -\sin x$

$\int \sin x \cdot (1-\cos^2 x) \cdot y^5 \frac{dy}{-\sin x}$

$-\int y^5 - y^7\, dy = -\frac{y^6}{6} + \frac{y^8}{8} + c$

$\frac{-\cos^6 x}{6} + \frac{\cos^8 x}{8}$

④ $\int_0^{\pi} \sin^2 x \cdot \cos^4 x\, dx$

$\int_0^{\pi} \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) \cdot \left(\frac{1}{2} + \frac{1}{2}\cos 2x\right)^2 dx$

$\int_0^{\pi} \left(\frac{1}{2} - \frac{1}{2}\cos 2x\right) \cdot \left(\frac{1}{4} + \frac{1}{2}\cos 2x + \frac{1}{4}\cos^2 2x\right)$

$\int_0^{\pi} \frac{1}{8} + \frac{1}{4}\cos 2x + \frac{1}{8}\cos^2 2x - \frac{1}{8}\cos 2x - \frac{1}{4}\cos^2 2x - \frac{1}{8}\cos^3 2x$

$\int_0^{\pi} \frac{1}{8}\left(1 + 2\cos 2x + \cos^2 2x - \cos 2x - 2\cos^2 2x - \cos^3 2x\right)$

$\int_0^{\pi} \frac{1}{8}(1 + \cos 2x - \cos^3 x) \cdot dx$

$\int_0^{\pi} \frac{1}{2}(1 + \cos 2x - (1-\sin^2 x)\cos 2x$ | $\sin x = y$, $\cos x\, dx = dy$

$\frac{1}{8} + \int \cos^3 2x \cdot dx$

$= \frac{1}{8} + \int_0^{\pi} \cos x (1-\sin^2 x) \frac{dy}{\cos x}$

$= \frac{1}{8} + y - \frac{y^3}{3}$

$= \frac{1}{8} - 1 + \frac{7}{8}$

Ans

$\int \tan^m x \cdot \sec^n x \, dx = ?$

① if $n = 2k$ (even), save a factor of $\sec^2 x$ and use $\sec^2 x = 1 + \tan^2 x$ to express the remaining factors in terms of $\tan x$ as:

$= \int \tan^m x \cdot \sec^{2k} x \, dx = \int \tan^m x \cdot \sec^{2k-2} x \cdot \sec^2 x \, dx$

$= \int \tan^m (1 + \tan^2 x) \cdot \sec^2 x \cdot dx$ | let $y = \tan x$

② if $m = 2k+1$ (odd) ... $\sec x \tan x$ ... $\tan^2 x = \sec^2 x - 1$ ... $\sec x$ ...

$\int \tan^{2k+1} x \cdot \sec^n x \, dx$

$= \int \tan^{2k} x \cdot \sec^{n-1} x \cdot \sec x \tan x \, dx$

$\int (\sec^2 x - 1)^k \cdot \sec^{n-1} x \cdot \sec x \tan x \, dx$

and let $y = \sec x$

$\cos x$ = Lā Ibās

$\cos(x+y) = \cos x \cos y - \sin x \sin y$

$\cos(x-y) = \cos x \cos y + \sin x \sin y$

$\cos x \cos y = \frac{1}{2}\cos(x+y) + \frac{1}{2}\cos(x-y)$

$\sin x$ = Lā Ibās

$\sin(x+y) = \sin x \cos y + \sin y \cos x$

$\sin(x-y) = \sin x \cos y - \sin y \cos x$

$\sin x \cos y = \frac{1}{2}\sin(x+y) + \frac{1}{2}\sin(x-y)$

$\sin x \sin y = \frac{1}{2}\cos(x-y) - \frac{1}{2}\cos(x+y)$

---

② $\int \tan x \cdot \sec x \cdot \sec^2 x \, dx$ | $y = \sec x$, $dy = \sec x \tan x$

$\int \tan x \cdot \sec x \cdot y^2 \, dy$

$\int y^2 \, dy = \frac{y^3}{3} = \frac{\sec^3 x}{3} + c$

② $\int \tan^5 x \cdot \sec^7 x \, dx$ ✓

③ $\int \tan^4 x \cdot \sec^6 x \, dx$ (6 even)

$= \int \tan^4 x \cdot \sec^4 x \cdot \sec^2 x \, dx$

$\int \tan^4 x \cdot (1+\tan^2 x)^2 \cdot \sec^2 x \, dx$ let $y = \tan x$, $dy = \sec^2 x$

$\int y^4 \cdot (1+y^2)^2 \cdot \sec^2 x \frac{dy}{\sec^2 x}$

$= \int y^4 \cdot (1 + 2y^2 + y^4) \, dx$

$= \int y^4 + 2y^6 + y^8 \cdot dx$

$= \frac{y^5}{5} + \frac{2y^7}{7} + \frac{y^9}{9} + c$

$= \frac{\tan^5 x}{5} + \frac{2}{7}\tan^7 x + \frac{\tan^9 x}{9} + c$

Amazi

7.4 Trigonometric Substitution

Expression	Substitution	Identity
$\sqrt{a^2-x^2}$	$x=a\sin\theta,\ \frac{-\pi}{2}\le\theta\le\frac{\pi}{2}$	$\cos^2\theta=1-\sin^2\theta$
$\sqrt{a^2+x^2}$	$x=a\tan\theta,\ \frac{-\pi}{2}<\theta<\frac{\pi}{2}$	$\sec^2x=1+\tan^2\theta$
$\sqrt{x^2-a^2}$	$x=a\sec\theta,\ 0\le\theta<\frac{\pi}{2}$ or $\pi\le\theta\le\frac{3\pi}{2}$	$\tan^2\theta=\sec^2\theta-1$

ex:- Evaluate the following:

① $\int \frac{\sqrt{9-x^2}}{x^2}\,dx$

Sol:- $a=3$

let $x=3\sin\theta$ $\quad \frac{dx}{d\theta}=3\cos\theta$

$\Rightarrow dx=3\cos\theta\,d\theta$

$\therefore \int \frac{\sqrt{9-9\sin^2\theta}}{9\sin^2\theta}\,3\cos\theta\,d\theta$

$\int \frac{9\cos^2\theta}{9\sin^2\theta}\,d\theta$

$=\int \cot^2\theta\,d\theta$

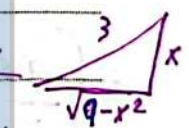
$=\int \csc^2\theta-1\,d\theta$

$=-\cot\theta-\theta+C$

$\sqrt{9-9\sin\theta}$	
$3\sqrt{1-\sin^2\theta}$	
$\cos^2\theta$	
$3\lvert\cos\theta\rvert$	

Since $x=3\sin\theta \Rightarrow \sin\theta=\frac{x}{3}$

$\theta=\sin^{-1}\left(\frac{x}{3}\right)$



$=\frac{-\sqrt{9-x^2}}{x}-\sin^{-1}\left(\frac{x}{3}\right)+C$

② $\int \frac{dx}{x^2\sqrt{x^2+4}}$

Sol: $a=2 \rightarrow x=2\tan\theta$

$\frac{dx}{d\theta}=2\sec^2\theta \Rightarrow dx=2\sec^2\theta\,d\theta$

$\int \frac{2\sec^2\theta\,d\theta}{4\tan^2\theta\cdot\sqrt{4\tan^2\theta+4}}$

$=\int \frac{2\sec^2\theta\,d\theta}{4\tan^2\theta\cdot 2\sec\theta}$

$=\int \frac{\sec\theta}{4\tan^2\theta}\,d\theta=\frac{1}{4}\int \frac{\frac{1}{\cos\theta}}{\frac{\sin^2\theta}{\cos^2\theta}}\,d\theta$

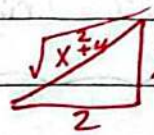
$=\frac{1}{4}\int \frac{\cos\theta}{\sin^2\theta}\,d\theta$

$\sqrt{4(1+\tan^2\theta)}$	
$\sqrt{4(1+\tan^2\theta)}$	
$\sqrt{4\sec^2\theta}$	
Amazing	
$2\lvert\sec\theta\rvert$	
$2\sec\theta$	

$\frac{1}{4}\int \frac{\cos\theta}{\sin\theta}\cdot\frac{1}{\sin\theta}\,d\theta$

$= \frac{1}{4}\int \cot\theta\csc\theta\,d\theta$

$= \frac{-1}{4}\csc\theta + C$ ⟶ $= \frac{-1}{4}\cdot\frac{\sqrt{x^2+4}}{x} + C$

since $\tan\theta = \frac{x}{2}$ ⟹ 

③ $\int \frac{dx}{\sqrt{x^2-a^2}}$, $a>0$

Sol:- let $x = a\sec\theta$

$\frac{dx}{d\theta} = a\sec\theta\tan\theta$

$dx = a\sec\theta\tan\theta\,d\theta$

$= \int \frac{a\sec\theta\cdot\tan\theta}{\sqrt{a^2\sec^2\theta - a^2}}\,d\theta$

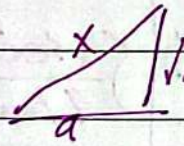
$= \int \frac{a\sec\theta\cdot\tan\theta}{a\tan\theta}\,d\theta$

$= \int \sec\theta\,d\theta$

$\times \frac{\sec+\tan}{\sec+\tan}$

$\sqrt{a^2\sec^2\theta - a^2}$	
$\sqrt{a^2(\sec^2\theta - 1)}$	
$\sqrt{a^2\tan^2\theta}$	
$= \lvert a\rvert\cdot\lvert\tan\theta\rvert$	
$= a\tan\theta$	$d\theta$

$= \ln|\sec\theta + \tan\theta| + C$

since $\sec\theta = \frac{x}{a}$ 

$\ln\left|\frac{x}{a} + \frac{\sqrt{x^2-a^2}}{a}\right| + C$

$\ln\left|\frac{x+\sqrt{x^2-a^2}}{a}\right| + C$

④ $\int \frac{x}{\sqrt{3-2x-x^2}}\,dx$

Sol: $3-2x-x^2 = 3-(2x+x^2)$

$= 3-(-1+1+2x+x^2)$

$= 3-(-1+(1+x)^2) = 4-(1+x)^2$

$= \int \frac{x}{4-(1+x)^2}\,dx$ $a=2$

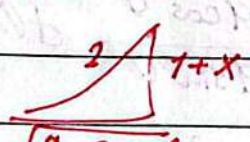
let $1+x = 2\sin\theta$ $\quad x = 2\sin\theta - 1$

$dx = 2\cos\theta\,d\theta$

$= \int \frac{2\sin\theta - 1}{\sqrt{4-4\sin^2\theta}}\cdot 2\cos\theta\,d\theta$

$= \int 2\sin\theta - 1\,d\theta$

$= -2\cos\theta - \theta + C$

$\sin\theta = \frac{1+x}{2}$ 

$= \frac{-2\sqrt{3-2x-x^2}}{2} - \sin^{-1}\left(\frac{1+x}{2}\right) + C$

Amazing

7.5 integrating Rational functions by Partial fractions.

① $$\frac{K}{(a_1x+b_1)(a_2x+b_2)\cdots(a_nx+b_n)} = \frac{A_1}{(a_1x+b_1)} + \frac{A_2}{(a_2x+b)} + \cdots + \frac{A_n}{a_nx+b_n}$$

② $$\frac{K}{(ax+b)^n} = \frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \cdots + \frac{A_n}{(ax+b)^n}$$

③ $$\frac{K}{(a_1x^2+b_1x+c_1)(a_2x^2+b_2x+c_2)\cdots(a_nx^2+b_nx+c_n)}$$
$$= \frac{A_1x+B_1}{a_1x^2+b_1x+c_2} + \frac{A_2x+B_2}{a_2x^2+b_2x+c_2} + \cdots + \frac{A_nx+B_n}{a_nx^2+b_nx+c_n}$$

④ $$\frac{K}{(ax^2+bx+c)^n} = \frac{A_1x+B_1}{ax^2+bx+c} + \frac{A_2x+B_2}{(ax^2+bx+c)^2} + \cdots + \frac{A_nx+B_n}{(ax^2+bx+c)^n}$$

الدرس الجديد

⑤ $$\int_0^3 \frac{dx}{x^2+9} = \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)\Big|_0^3 = \frac{\pi}{12}$$

⑥ $$= \int_0^{\frac{\pi}{4}} \frac{3\sec^2\theta}{9\sec^2\theta} = \int_0^{\frac{\pi}{4}} \frac{1}{3}\,\theta = \frac{\pi}{12}$$

Sol: $a=3 \rightarrow x = 3\tan\theta$

$\boxed{dx = 3\sec^2\theta\, d\theta}$

$x=0 \rightarrow 0 = 3\tan\theta \rightarrow \tan\theta = 0 \rightarrow \theta = 0$

$x=3 \rightarrow 3 = 3\tan\theta \rightarrow \tan\theta = 1 \rightarrow \theta = \frac{\pi}{4}$

$$= \int_0^{\frac{\pi}{4}} \frac{3\sec^2\theta}{9\tan^2\theta+9}\, d\theta$$

$9(\tan^2\theta+1) = 9\sec^2\theta$

(Note that $\theta = \tan^{-1}\left(\frac{x}{3}\right)$)

Amazing

Ex: Write out the form of the partial fraction decomposition of the function:-

$$\frac{x^3+x^2+1}{x(x-1)(x^2+x+1)(x^2+1)^2} = \frac{A_1}{x} + \frac{A_2}{x-1} + \frac{A_3x+A_4}{x^2+x+1} + \frac{A_5x+A_6}{x^2+1} + \frac{A_7x+A_8}{(x^2+1)^2}$$

$B^2-4AC<0$

① $\int \frac{dx}{x^2+5x+6}$

Sol:-

$$\frac{1}{x^2+5x+6} \Rightarrow \frac{1}{(x+3)(x+2)} \rightarrow \frac{A}{x+3} + \frac{B}{x+2} = \frac{A(x+2)+B(x+3)}{(x+2)(x+3)}$$

$$A(x+2)+B(x+3)=1$$

$x=-2 \longrightarrow B(-2+3)=1 \Rightarrow B=1$

$x=-3 \longrightarrow A(-3+2)=1 \Rightarrow A=-1$

$$\int \frac{-1}{x+3} + \frac{1}{x+2}\,dx$$

$$= -\ln|x+3| + \ln|x+2| + C$$

$$= \ln\left|\frac{x+2}{x+3}\right| + C$$

$$\frac{x^3+x^2+1}{x(x-1)(x^2+x+1)(x^2+1)^2} = \frac{A_1}{x} + \frac{A_2}{x-1} + \frac{A_3x+A_4}{x^2+x+1} + \frac{A_5x+A_6}{x^2+1} + \frac{A_7x+A_8}{(x^2+1)^2}$$

$B^2-4AC<0$

Am

$$\int \frac{dx}{x^3+4x}$$

$$\frac{1}{x^3+4x} = \frac{1}{x\cdot(x^2+4)} = \frac{A}{x} + \frac{Bx+C}{x^2+4}$$

$$= \frac{A(x^2+4)+(Bx+C)x}{x(x^2+4)}$$

$$A(x^2+4)+(Bx+C)x = 1$$

$$A(x^2+4)+Bx^2+Cx = 1$$

$$Ax^2+4A+Bx^2+Cx = 1$$

$$(A+B)x^2+4A+Cx = 1$$

$4A = 1 \quad A = \frac{1}{4}$

$C = 0$ and $A+B = 0 \quad B = -A = -\frac{1}{4}$

$$\int \left(\frac{\frac{1}{4}}{x} + \frac{-\frac{1}{4}x}{x^2+4} \right) dx$$

$$= \frac{1}{4}\int \frac{1}{x}dx - \frac{1}{4}\cdot\frac{1}{2}\int \frac{2x}{x^2+4}dx$$

$$= \frac{1}{4}\ln|x| - \frac{1}{8}\ln|x^2+4| + C$$

③ $$\int \frac{x^2+2x-1}{2x^3+3x^2-2x}dx$$

sol: $2x^3+3x^2-2x \Rightarrow x(2x^2+3x-2)$

$\Rightarrow x(2x-1)(x+2)$

$$\frac{x^2+2x-1}{x(2x-1)(x+2)} = \frac{A}{x} + \frac{B}{2x-1} + \frac{C}{(x+2)}$$

$$\frac{A(2x-1)(x+2)+Bx(x+2)+Cx(2x-1)}{x(2x-1)(x+2)}$$

$$A(2x-1)(x+2)+Bx(x+2)+Cx(2x-1) = x^2+2x-1$$

$x=0 \rightarrow A\cdot(-1)\cdot 2+0+0 = -1 \quad A = \frac{1}{2}$

$x=\frac{1}{2} \rightarrow 0+\frac{5}{4}B+0 = \frac{1}{4} \quad B = \frac{1}{5}$

$x=-2 \rightarrow 0+0+10C = -1 \quad C = \frac{-1}{10}$

$$= \int \left(\frac{\frac{1}{2}}{x} + \frac{\frac{1}{5}}{2x-1} - \frac{\frac{1}{10}}{x+2} \right) dx$$

$$= \frac{1}{2}\ln|x| + \frac{1}{5}\cdot\frac{1}{2}\ln|2x-1| - \frac{1}{10}\ln|x+2| +$$

Amazing

④ $\int \frac{x^3+x+1}{x^2-1}\,dx$

$$\begin{array}{r|l} x^3+x+1 & x^2-1 \\ -(x^3-x) & x \\ \hline 2x+1 & \end{array}$$

$$= \int \left(x + \frac{2x+1}{x^2+1}\right) dx$$

$$= \frac{x^2}{2} \int \frac{2x+1}{x^2-1}\,dx$$

⑤ $\int \frac{\sqrt{x+4}}{x}\,dx$

sol:- Let $y = \sqrt{x+4} \rightarrow y^2 = x+4$

$dy = \frac{1}{2\sqrt{x+4}}$ $\quad x = y^2 - 4$

$$= \int \frac{y}{y^2-4} \cdot 2y\,dy$$

$$= \int \frac{2y^2}{y^2-4}\,dy$$

$$\begin{array}{r|l} 2y^2 & y^2-4 \\ -(2y^2-8) & 2 \\ \hline 8 & \end{array}$$

$$\int 2 + \frac{8}{y^2-4}\,dy$$

$$= 2y \int \frac{8}{y^2-4}\,dy$$

⑥ $\int \frac{\cos x}{\sin^2 x + 5\sin x + 4}\,dx$

⑦ $\int \frac{1 - x + 2x^2 - x^3}{x(x^2+1)^2}$

للحل ← ∞

7, 8. Improper Integral

Type ① ∞ Infinite series

① $\int_a^{\infty} f(x)\,dx = \lim_{b \to \infty} \int_a^b f(x)\,dx$

divergent if the limit doesn't exist / convergent if the limit exists

d.n.e

② $\int_{-\infty}^{a} f(x)\,dx = \lim_{b \to -\infty} \int_b^a dx$

③ $\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{a} f(x)\,dx + \int_a^{\infty} f(x)\,dx$

(a: Any real number and f is cont. at a)

Amaz

Ex: Evaluate the following

① $\int_1^\infty \frac{1}{x}\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx$

$= \lim_{b\to\infty} [\ln|x|]_1^b$

$= \lim_{b\to\infty} (\ln|b| - 0) = \infty \in \mathbb{R}$

Div..

② $\int_{-\infty}^0 x\cdot e^x\,dx = \lim_{b\to-\infty} \int_b^0 x\cdot e^x\,dx$

+	x	e^x
−	1	e^x
±	0	e^x

$= \lim_{b\to-\infty} \left((x-1)e^x\right|_b^0$

$= \lim_{b\to-\infty} (-1 - (b-1)e^b)$

$= -1 - \lim_{b\to-\infty} \frac{b-1}{e^{-b}} = -1 - \lim_{b\to-\infty} \frac{1}{-e^{-b}}$

$= -1 + \frac{1}{\infty} = 0$

$= -1$

③ $\int_1^\infty \frac{x^2}{\sqrt{x^3+1}}\,dx = \lim_{b\to\infty} \int_1^b \frac{x^2}{\sqrt{x^3+1}}\,dx$

Let $y = x^3+1 \Rightarrow dx = \frac{dy}{3x^2}$

$= \lim_{b\to\infty} \int_{\square}^{\square} \frac{x^2}{\sqrt{y}} \frac{dy}{3x^2} = \lim_{b\to\infty} \frac{1}{3} \int_{\square}^{\square} y^{\frac{1}{2}}\,dy$

$= \lim_{b\to\infty} \frac{2}{3}\sqrt{y}\Big|_{\square}^{\square} = \lim_{b\to\infty} \frac{2}{3}\sqrt{x^3+1}\Big|_1^b$

$\lim_{b\to\infty} \left(\frac{2}{3}\sqrt{b^3+1} - \frac{2}{3}\sqrt{2}\right)$

$= \infty - \frac{2}{3}\sqrt{2} = \infty$ "div."

④ $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$

Sol: $f(x) = \frac{1}{1+x^2}$ is cont. at $x=0$

$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \int_{-\infty}^{0} \frac{dx}{1+x^2} + \int_0^{\infty} \frac{dx}{1+x^2}$

$\int_{-\infty}^{0} \frac{dx}{1+x^2} = \lim_{b\to-\infty} \int_b^0 \frac{dx}{1+x^2}$

$= \lim_{b\to-\infty} \tan^{-1}(x)\Big|_b^0 = \lim_{b\to-\infty} (0 - \tan^{-1}(b))$

$= -\tan^{-1}(-\infty) = \tan^{-1}(\infty) = \frac{\pi}{2}$ "Conv."

$\int_0^{\infty} \frac{dx}{1+x^2} = \lim_{b\to\infty} \int_0^b \frac{dx}{1+x^2} = \lim_{b\to\infty} \tan^{-1}(x)\Big|_0^b = \tan^{-1}(\infty) = \frac{\pi}{2}$

$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$

Amazing

Type II: Discontinuous Integrals

① if f is cont. on $[a,b)$ and is discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$$

- → convergent if the limit exists
- → divergent if the limit does not exist

② " " " " " $(a,b]$ " " " at a, then

$$\int_a^b f(x) = \lim_{t\to a^+} \int_t^b f(x)\,dx$$

- → convergent if the limit exists
- → divergent if the limit does not exi

③ if f has a discontinuity point at c where $a<c<b$ and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

$$= \lim_{t\to c^-} \int_a^t f(x)\,dx + \lim_{t\to c^+} \int_t^b f(x)\,dx.$$

Ex: Evaluate the following:-

① $\int_2^6 \frac{dx}{\sqrt{x-2}}$

Sol:- $f(x) = \frac{1}{\sqrt{x-2}}$ discont. at $x=2$

$= \lim_{t\to 2^+} \int_t^6 (x-2)^{-\frac{1}{2}} dx = \lim_{t\to 2^+} 2\sqrt{x-2}\Big|_t^6$

$= \lim_{t\to 2^+} (4 - 2\sqrt{t-2})$

$= 4 - 2 \cdot \lim_{t\to 2^+} \sqrt{t-2}$ $\quad t-2=0 \Rightarrow t=2$

$4 - 2\sqrt{0}$ "conv." $\quad - - - \overset{2}{0} + + +$

$= 4$

② $\int_0^{\frac{\pi}{2}} \sec x\, dx$

Solution $f(x) = \sec x = \frac{1}{\cos x}$

discont at $x = \frac{\pi}{2}$

$\lim_{t\to \frac{\pi}{2}^-} \int_0^t \sec x\, dx$

$= \lim_{t\to \frac{\pi}{2}^-} \ln|\sec x + \tan x|_0^t$

$= \lim_{t\to \frac{\pi}{2}^-} \ln\left|\frac{1+\sin x}{\cos x}\right|_0^t$

$= \lim_{t\to \frac{\pi}{2}^-} \left(\ln\left|\frac{1+\sin t}{\cos t}\right| - \ln(1)\right)$ ($\ln(1) = 0$)

$= \ln(\infty) = \infty$ div

③ $\int_0^1 \ln x\, dx$

$= \lim_{t\to 0^+} \int_t^1 \ln x\, dx$

$u = \ln x \longrightarrow du = \frac{1}{x} dx$

$dv = dx \longrightarrow v = x$

$= \lim_{t\to 0^+} \left(x \cdot \ln x - \int 1\, dx\right)$

$= \lim_{t\to 0^+} (x \cdot \ln x - x)\Big|_t^1$

$= \lim_{t\to 0^+} \left((0-1) - (t \cdot \ln t - t)\right)$

$= -1 - \lim_{t\to 0^+} (t \ln t - t)$ $\quad t(\ln t - 1)$

$= -1 - \lim_{t\to 0^+} \frac{\ln t - 1}{\frac{1}{t}}$ $\quad \frac{-\infty}{\infty}$ L'Hôpital

$= -1 - \lim_{t\to 0^+} \frac{\frac{1}{t}}{\frac{-1}{t^2}}$

$= -1 - \lim_{t\to 0^+} (-t) = -1 - 0 = \boxed{-1}$ "conv."

(42) $\int_0^3 \frac{dx}{x-1}$ Sol: $f(x)=\frac{1}{x-1}$ discont at $x=1$ and $1 \in [0,3]$

$$\int_0^3 \frac{dx}{x-1} = \int_0^1 \frac{dx}{x-1} + \int_1^3 \frac{dx}{x-1}$$

$$\int_0^1 \frac{dx}{x-1} = \lim_{t\to 1^-} \int_0^t \frac{dx}{x-1}$$

$$= \lim_{t\to 1^-} \ln|x-1|\Big|_0^t = \lim_{t\to 1^-} (\ln|t-1| - \ln|1|) = \lim_{t\to 1^-} \ln(1-t)$$

let $1-t=z$

$t\to 1^- \;/\; z\to 0^+$

$\lim_{z\to 0^+} \ln z = -\infty$ "div"

So $\int_0^3 \frac{dx}{x-1}$ "div"

Chapter ⑨: Infinite Series

9.1 Sequences

Def: A sequence is a function $f(n)=a_n$, $n=1,2,3,\ldots$ or $\{a_n\}_{n=1}^{\infty} = \{a_1, a_2, a_3, \ldots\}$ whose domain is a set of positive integer numbers.

Ex: In each part, find the general term of the following sequences:-

① $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \ldots$

$a_1=\frac{1}{2}=\frac{1}{1+1}$

$a_2=\frac{2}{3}=\frac{2}{2+1}$

$a_3=\frac{3}{4}=\frac{3}{3+1}$

$a_n=\boxed{\frac{n}{n+1}}$, $n=1,2,3,\ldots$

② $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \ldots$

$a_1=\frac{1}{2}=\frac{1}{2}$

$a_2=\frac{1}{4}=\frac{1}{2^2}$

$a_3=\frac{1}{8}=\frac{1}{2^3}$

$a_n=\frac{1}{2^n}=\left(\frac{1}{2}\right)^n$

Amazing

③ $1, -3, 5, -7, 9, -11, \ldots$

$a_1 = 1 = 2(1)-1$

$a_2 = -3 = -(2(2)-1)$

$a_3 = 5 = (2(3)-1)$

$a_4 = -7 = -(2(4)-1)$

$\vdots$

$a_n = (-1)^{n+1}(2n-1)$

Def: A seq $\{a_n\}_{n=1}^{\infty}$ is said to be converge to $L \in \mathbb{R}$ if

$\lim_{n\to\infty} a_n = L$, other wise the sequence is said to be diverge

---

Theorem: if $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are convergent and c is a constant, then:-

① $\lim_{n\to\infty}(c) = c$

② $\lim_{n\to\infty}(c \cdot a_n) = c \cdot \lim_{n\to\infty} a_n$

③ $\lim_{n\to\infty}(a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n$

④ $\lim_{n\to\infty}(a_n \cdot b_n) = \lim_{n\to\infty} a_n \cdot \lim_{n\to\infty} b_n$

⑤ $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$ if $\boxed{\lim_{n\to\infty} b_n \neq 0}$

Ex: Determine wether the sequence conv. or div:

① $\left\{\frac{n}{2n+1}\right\}_{n=1}^{\infty}$

Sol: $a_n = \frac{n}{2n+1} \Rightarrow \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2}$

$\Rightarrow$ The seq conv. and $\left\{\frac{n}{2n+1}\right\}_{n=1}^{\infty} = \left\{\frac{1}{3}, \frac{2}{5}, \frac{3}{7}, \ldots\right\} \to \frac{1}{2}$

② $\left\{(-1)^{n+1}\cdot\frac{n}{2n+1}\right\}_{n=1}^{\infty}$

Sol: $\lim_{n\to\infty}(-1)^{n+1}\frac{n}{2n+1} = \lim_{n\to\infty}(-1)^{n+1}\cdot\lim_{n\to\infty}\frac{n}{2n+1}$ $= \frac{1}{2}$

$= \frac{1}{2}\cdot\lim_{n\to\infty}(-1)^{n+1} = \frac{1}{2}\cdot\begin{cases}1, & \text{if } n: \text{odd}\\ -1, & \text{if } n: \text{even}\end{cases}$

$= \begin{cases}\frac{1}{2}, & \text{if } n: \text{odd}\\ -\frac{1}{2}, & \text{if } n: \text{even}\end{cases}$

$\Rightarrow \lim_{n\to\infty}(-1)^{n+1}\frac{n}{2n+1}$ dose not exiest

"div"

③ $\{8-2n\}_{n=1}^{\infty}$

Sol: $\lim_{n\to\infty}(8-2n) = \lim_{n\to\infty}(-2n) = -\infty \notin \mathbb{R}$

"Div"

Amazing

④ $\left\{\frac{n^3+n+1}{n^2-n+2}\right\}_{n=1}^{\infty}$

Sol: $\lim_{n\to\infty}\frac{n^3+n+1}{n^2-n+2}=\lim_{n\to\infty}\frac{n^3}{n^2}$

$=\lim_{n\to\infty} n=\infty \Rightarrow$ "div"

⑤ $\left\{\frac{n}{e^n}\right\}_{n=1}^{\infty}$

Sol: $\lim_{n\to\infty}\frac{n}{e^n}=\lim_{n\to\infty}\frac{1}{e^n}=\frac{1}{e^\infty}=\frac{1}{\infty}=0$

"Conv" $=0$ ✓

⑥ $\left\{\sqrt{n^2+3n}-n\right\}_{n=1}^{\infty}$

Sol: $\lim_{n\to\infty}\sqrt{n^2+3n}-n\cdot\frac{\sqrt{n^2+3n}+n}{\sqrt{n^2+3n}+n}$

$\lim_{n\to\infty}\frac{n^2+3n-n^2}{\sqrt{n^2+3n}+n}=\lim_{n\to\infty}\frac{3n}{\sqrt{n^2+n}}$

$=\lim_{n\to\infty}\frac{3n}{|n|+n}=\frac{3n}{2n}=\frac{3}{2}\Rightarrow$ Conv

⑦ $\left\{\left(1-\frac{2}{n}\right)^n\right\}_{n=1}^{\infty}$

Sol: $\lim_{n\to\infty}\left(1-\frac{2}{n}\right)^n=\left(1-\frac{2}{\infty}\right)^{\infty}=(1-0)^{\infty}$

$=1^{\infty}$

$\lim_{n\to\infty}\left(1-\frac{2}{n}\right)^n=e^{\lim_{n\to\infty}\frac{\ln\left(1-\frac{2}{n}\right)}{\frac{1}{n}}}$

$\lim_{n\to\infty}\frac{\frac{2}{n^2}}{\frac{1-\frac{2}{n}}{\left(\frac{1}{n^2}\right)}}=e^{\lim_{n\to\infty}\frac{2}{n^2\left(1-\frac{2}{n}\right)}\cdot\frac{n^2}{1}}$

$e^{\lim_{n\to\infty}\frac{-2}{1-\frac{2}{n}}}=e^{\frac{-2}{1-0}}=e^{-2}\Rightarrow$ "Conv"

⑧ $\left\{\frac{\sin^2(n)}{n+1}\right\}_{n=1}^{\infty}$

Sol: By squeeze Theorem

$-1\le\sin(n)\le 1$

$0\le\sin^2(n)\le 1$

$0\le\frac{\sin^2(n)}{n+1}\le\frac{1}{n+1}$

$\lim_{n\to\infty}(0)=0=\lim_{n\to\infty}\frac{1}{n+1}=\frac{1}{\infty}=0$

$\Rightarrow\lim_{n\to\infty}\frac{\sin^2(n)}{n+1}=0\Rightarrow$ "Conv"

Amazing

9.2. Infinite Series

Def: An infinite series is an expression that can be written in the form:

$$\sum_{k=1}^{\infty} u_k = u_1 + u_2 + \cdots + u_k + \cdots \quad (1)$$

The numbers $u_1, u_2, \ldots$ are called the terms of the series

(*) Let $S_1 = u_1$ / $S_2 = u_1 + u_2$

$S_3 = u_1 + u_2 + u_3$

$\vdots$

$S_n = u_1 + u_2 + \cdots + u_n = \sum_{k=1}^{n} u_k$

Then $\{S_n\}_{n=1}^{\infty} = \{S_1, S_2, \ldots\}$ is called the sequence of the partial sums

Theorem (Test (1)): ① if the seq. of the partial sums $\{S_n\}_{n=1}^{\infty}$ [illegible] convergent and $\{S_n\}_{n=1}^{\infty} \to$ conv $\to S$ $(\lim_{n\to\infty} S_n = S)$

Then the series

① is convergat and $\sum_{k=1}^{\infty} u_k = S$.

② if the seq. of the partial sums $\{S_n\}_{n=1}^{\infty}$ is divergent, then the series (1) is divergent

Ex: Determine wether the series converges, and if so, find it's sum

① $\sum_{k=1}^{\infty} \left(\frac{1}{k+3} - \frac{1}{k+4}\right)$

Sol: $S_n = u_1 + u_2 + u_3 + \ldots + u_{n-2} + u_{n-1} + u_n$

$= \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) + \cdots$

$+ \left(\frac{1}{n+1} - \frac{1}{n+2}\right) + \left(\frac{1}{n+2} - \frac{1}{n+3}\right) + \cdots$

$S_n = \frac{1}{4} - \frac{1}{n+4} \Rightarrow \left\{\frac{1}{4} - \frac{1}{n+4}\right\}_{n=1}^{\infty}$

$\lim_{n\to\infty}\left(\frac{1}{4} - \frac{1}{n+4}\right) = \frac{1}{4} - \frac{1}{\infty} = \frac{1}{4} - 0 = \frac{1}{4}$

$\Rightarrow \left\{\frac{1}{4} - \frac{1}{n+4}\right\}_{n=1}^{\infty}$ conv $= \frac{1}{4}$

So $\sum_{k=1}^{\infty}\left(\frac{1}{k+3} - \frac{1}{k+4}\right)$ conv $= \frac{1}{4}$

Amazing

② $\sum_{k=1}^{\infty} \frac{2}{k(k+2)}$

Sol:- $\frac{2}{k(k+2)} = \frac{A}{k} + \frac{B}{k+2} = \frac{A(k+2)+Bk}{k(k+2)}$

$A(k+2)+Bk = 2$

$k=0 \Rightarrow 2A = 2 \Rightarrow \boxed{A=1}$

$k=-2 \Rightarrow -2B = 2 \Rightarrow \boxed{B=-1}$

$\frac{2}{k(k+2)} = \frac{1}{k} - \frac{1}{k+2} = U_k$

$S_n = U_1+U_2+U_3+U_4+\dots+U_{n-3}+U_{n-3}+U_{n-2}+U_{n-1}+U_n$

$= \left(1-\frac{1}{3}\right)+\left(\frac{1}{2}-\frac{1}{4}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)+\left(\frac{1}{4}-\frac{1}{6}\right)+\dots\dots$

$\left(\frac{1}{n-3}-\frac{1}{n-1}\right)+\left(\frac{1}{n-2}-\frac{1}{n}\right)+\left(\frac{1}{n-1}-\frac{1}{n+1}\right)+\left(\frac{1}{n}-\frac{1}{n+2}\right)$

$1+\frac{1}{2}-\frac{1}{n+1}-\frac{1}{n+2} = \frac{3}{2}-\frac{1}{n+1}-\frac{1}{n+2}$

$= \left\{\frac{3}{2}-\frac{1}{n+1}-\frac{1}{n+2}\right\}_{n=1}^{\infty} \Rightarrow \lim_{n\to\infty} \frac{3}{2}-\frac{1}{n+1}-\frac{1}{n+2} = \frac{3}{2}-0-0$

$\Rightarrow \left\{\frac{3}{2}-\frac{1}{n+1}-\frac{1}{n+2}\right\}_{n=1}^{\infty}$ conv. and $= \frac{3}{2}$

So $\sum_{k=1}^{\infty} \frac{k}{k(k+2)}$ conv and $= \frac{3}{2}$

③ $\sum_{k=2}^{\infty} \frac{1}{k^2-1}$ ④ $\sum_{k=1}^{\infty} \frac{1}{k^2+5k+6}$

⑤ $\sum_{k=1}^{\infty} \left(\tan^{-1}(k) - \tan^{-1}(k+1)\right)$

Theorem (Test 2): (Geometric series):
The geometric series $\sum_{k=0}^{\infty} a r^k = a + ar + ar^2 + \dots = \frac{a}{1-r}$ if the series convergent
it's converg. if $|r|<1$ and diverges if $|r| \geq 1$

Ex: Determine wether the series converges, and if so, find it's sum

① $1-1+1-1+\dots$

Sol: $r = \frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3} = \dots = \frac{-1}{1} = -1$

$|r| = |-1| = 1 \Rightarrow$ Div.

② $\sum_{k=0}^{\infty} \frac{5}{4^k} = \sum_{k=0}^{\infty} 5\left(\frac{1}{4}\right)^k$

$r = \frac{1}{4} \Rightarrow |r| = \frac{1}{4} < 1 \Rightarrow$ conv.

So, $\sum_{k=0}^{\infty} \frac{5}{4^k} = \frac{5}{1-\frac{1}{4}} = \frac{5}{\frac{3}{4}} = \frac{20}{3}$

Ama

③ $\sum_{k=1}^{\infty}\left(\frac{2}{3}\right)^{k+2}=\sum_{k=1}^{\infty}\underbrace{\left(\frac{2}{3}\right)^{3}}_{a}\cdot\left(\frac{2}{3}\right)^{k-1}$

$r=\frac{2}{3}\Rightarrow|r|=\frac{2}{3}<1\Rightarrow$ conv.

$\sum_{k=1}^{\infty}\left(\frac{2}{3}\right)^{k+2}=\frac{\left(\frac{2}{3}\right)^{3}}{1-\frac{2}{3}}=\frac{\left(\frac{2}{3}\right)^{3}}{\frac{1}{3}}=\frac{8}{9}$

④ $\sum_{k=1}^{\infty}3^{2k}\cdot5^{1-k}=\sum_{k=1}^{\infty}9^{k}\cdot\frac{5}{5^{k}}=\sum_{k=1}^{\infty}5\left(\frac{9}{5}\right)^{k}$

$=\sum_{k=1}^{\infty}5\cdot\left(\frac{9}{5}\right)\left(\frac{9}{5}\right)^{k-1}=\sum_{k=1}^{\infty}9\cdot\left(\frac{9}{5}\right)^{k-1}$

$|r|=\left|\frac{9}{5}\right|=\frac{9}{5}>1\Rightarrow$ Div.

⑤ $\sum_{k=1}^{\infty}(-1)^{k-1}\cdot\frac{7}{6^{k+1}}=\sum_{k=1}^{\infty}\frac{7}{6^{2}}\cdot\left(\frac{-1}{6}\right)^{k-1}$

$|r|=\left|\frac{-1}{6}\right|=\frac{1}{6}<1\Rightarrow$ Conv.

$\sum_{k=1}^{\infty}(-1)^{k-1}\cdot\frac{7}{6^{k+1}}=\frac{\frac{7}{6^{2}}}{1-\frac{-1}{6}}=\frac{7}{6^{2}}\cdot\frac{6}{7}=\frac{1}{6}$

⑥ $0.784784784\ldots=0{,}784+0{,}000784+\ldots$

$r=\frac{0{,}000784}{0{,}784}=\frac{1}{1000}=0{,}001$

$|r|=0{,}001<1\Rightarrow$ conv

$\frac{a}{\infty}=\frac{0{,}784}{1-0{,}001}=\frac{784}{999}$

⑧ $\sum_{k=1}^{\infty}\frac{4^{k+2}}{7^{k-1}}$ ⑨ $\sum_{k=5}^{\infty}\left(\frac{e}{\pi}\right)^{k-1}$

Ex: Find all values of x for which the series converges, and find the sum of the series for those values of x

① $x-x^{3}+x^{5}-x^{7}+\ldots$

② $\frac{1}{x^{2}}+\frac{2}{x^{3}}+\frac{4}{x^{4}}+\frac{8}{x^{5}}+\ldots\rightarrow\frac{2^{2}}{x^{2}}\ \frac{2^{3}}{x^{3}}$

$\frac{2^{n}}{x^{2+n}}$

③ $e^{-x}+e^{-2x}+e^{-3x}+\ldots\rightarrow$ div

$e^{-x\cdot n}$

Sol: ① $r=\frac{-x^{3}}{x}=-x^{2}\Rightarrow|r|=|-x^{2}|=x^{2}$

The series conv if $x^{2}<1\Rightarrow-1<x<1$

So, the series conv. on $(-1,1)$

$\text{Sum}=\frac{x}{1+x^{2}}$

Amazing

9.4 Convergence Tests

Theorem (The divergence Test (3))

① if $\lim_{k\to\infty} u_k \neq 0$, then the series $\sum u_k$ diverges.

② If $\lim_{k\to\infty} u_k = 0$, then the test fails

Ex: Determine wether the series converges or diverges

① $\sum_{k=1}^{\infty} \frac{k^3-k+1}{k^2+k}$

Sol: $\lim_{k\to\infty} \frac{k^3-k+1}{k^2+k} = \lim_{k\to\infty} \frac{k^3}{k^2} = \lim_{k\to\infty} k = \infty \neq 0$

The series div.

② $\sum_{n=1}^{\infty} \frac{n+e^n}{e^n+1}$

Sol: $\lim_{n\to\infty} \frac{n+e^n}{e^n+1} = \frac{e^n}{e^n} = 1 \neq 0$

The series div

③ $\sum_{k=1} \tan^{-1}(k)$

Sol: $\lim_{k\to\infty} \tan^{-1}(k) = \tan^{-1}(\infty) = \frac{\pi}{2} \neq 0$ "Div"

④ $\sum_{k=1}^{\infty} \sqrt[k]{e} = \lim_{k\to\infty} e^{\frac{1}{k}} = e^{\frac{1}{\infty}} = e^0 = 1 \neq 0$ "div"

⑤ $\sum_{k=1}^{\infty} \left(1+\frac{1}{k}\right)^{-k}$

⑥ $\sum_{k=1}^{\infty} \frac{k}{\ln(k+1)}$

Theorem (The Integral Test (4))

Let $\sum u_k$ be a series with positive terms, and let $f(x)$ be the general term of the series, if f is continuous and decreasing on $[a,\infty)$, then $\sum_{k=1}^{\infty} u_k$ and $\int_a^{\infty} f(x)\,dx$ both converge or diverge.

Ex: Determine wether the series converges or diverges:

① $\sum_{k=1}^{\infty} \frac{k}{k^2+1}$

Sol: $u_k = \frac{k}{k^2+1} \Rightarrow f(x) = \frac{x}{x^2+1}$

$f'(x) = \frac{(x^2+1)-(x)(2x)}{(x^2+1)^2} = \frac{1-x^2}{(x^2+1)^2}$ ↗ ∞ ↘ 1 ↗

$1-x^2=0 \Rightarrow \boxed{x=\pm 1}$

Sign(f'): − − − | + + + + | − − − →, with −1 and 1

$\Rightarrow a=1$

$\Rightarrow f$ is cont & decreasing $[1,\infty)$

$$\int_1^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_1^b \frac{2x}{x^2+1}\,dx$$

$$= \lim_{b\to\infty} \frac{1}{2}\ln|x^2+1|\Big|_1^b$$

$$= \lim_{b\to\infty} \frac{1}{2}\left(\ln|b^2+1| - \ln|2|\right)$$

$$= \frac{1}{2}(\infty - \ln 2) = \infty \text{ div}$$

$\Rightarrow$ The series div.

Amazing

② $\sum_{k=1}^{\infty} \frac{1}{1+9k^2}$

Sol $f(x)=\frac{1}{1+9x^2} \to f'(x)=\frac{-18x}{(1+9x^2)^2}=0$

$-18x=0 \longrightarrow x=0$



f is cont & dec. on $[0,\infty)$

$$\int_0^{\infty} \frac{1}{1+9x^2}dx = \lim_{b\to\infty} \int_0^b \frac{1}{1+(3x)^2}dx$$

$$\lim_{b\to\infty} \int_0^b \frac{1}{1+y^2}\frac{dy}{3} = \frac{1}{3}\left(\tan^{-1}(3b) - \tan^{-1}(0)\right)$$

Let $y=3x$, $dy=3\,dx$

$\frac{1}{3}\tan^{-1}(\infty)=\frac{1}{3}\cdot\frac{\pi}{2}=\frac{\pi}{6}$

"conv."

③ $\sum_{k=1}^{\infty} \frac{1}{k}$

Sol: $f(x)=\frac{1}{x} \Rightarrow f'(x)=\frac{-1}{x^2}\neq 0$

But $x^2=0 \longrightarrow x=0$



So, $\int_1^{\infty} f(x)dx \overset{\lim}{=} \int_1^b \frac{1}{x}dx$

$= \lim_{b\to\infty} \ln|x|\Big|_1^b = \lim_{b\to\infty} (\ln|b| - \underbrace{\ln|1|}_{\text{zero}}) = \infty$

"div"

④ $\sum_{k=3}^{\infty} \frac{\ln(|k|)}{k}$ ⑤ $\sum_{n=1}^{\infty} \frac{1}{n^2}$

⑥ $\sum_{k=1}^{\infty} k^2 \cdot e^{-|k^3|}$ ⑦ $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3+1}}$

⑧ $\sum_{k=1}^{\infty} \frac{\tan^{-1}(k)}{k^2+1}$

Theorem (The convergence of p-series (Test ③)):

$\sum_{k=1}^{\infty} \frac{1}{k^p} = 1+\frac{1}{2^p}+\frac{1}{3^p}+\cdots$ converges if $p>1$ and diverges if $0<p\leq 1$

Remark: if $p<0$, then by divergence test the series is divergent.

Theorem: if $\sum u_k$ and $\sum v_k$ are converged series and $c\neq 0$, then

① $\sum(u_k \pm v_k)$ is convergent and $\sum(u_k \pm v_k)=\sum u_k \pm \sum v_k$

② $\sum c\cdot u_k$ is convergent and $\sum c\cdot u_k = c\sum u_k$

Remark: if $\sum u_k$ is divergent and $(c\neq 0)$ then $\sum c\cdot u_k$ is divergent

Amazing

Ex: Determine wether the series converges.

① $\sum_{k=1}^{\infty} \frac{1}{\sqrt[3]{k^2}} = \sum_{k=1}^{\infty} \frac{1}{k^{\frac{2}{3}}}$ div $P=\frac{2}{3}<1$

② $\sum_{k=1}^{\infty} \frac{5}{k^3}$ = Sol: Since $\sum_{k=1}^{\infty} \frac{1}{k^3}$ conv $P=3>1$

Then $\sum_{k=1}^{\infty} \frac{5}{k^3}$ 'conv'

③ $\sum_{k=1}^{\infty} \frac{3k^2-1}{k^4} = \sum_{k=1}^{\infty} \left(\frac{3}{k^2} - \frac{1}{k^4}\right)$ (u_k, v_k)

Since $\sum_{k=1}^{\infty} \frac{1}{k^2}$ "conv" Then $\sum_{k=1}^{\infty} \frac{3}{k^2}$ "conv"

and $\sum_{k=1}^{\infty} \frac{1}{k^4}$ conv So: $\sum_{k=1}^{\infty} \frac{3k^2-1}{k^4}$ Conv.

Q: Determine wether the series

$\sum_{k=1}^{\infty} \left(\frac{2}{(-5)^k} + \frac{1}{k} - \frac{1}{k+1}\right)$ (u_k, v_k)

if so, find it's sum

Sol: $\sum_{k=1}^{\infty} \frac{2}{(-5)^k} = \sum_{k=1}^{\infty} 2\left(\frac{1}{-5}\right)^k = \sum_{k=1}^{\infty} \frac{-2}{5} \cdot \left(\frac{-1}{5}\right)^{k-1}$

By geometric series

$|r| = \left|\frac{-1}{5}\right| = \frac{1}{5} < 1$ "conv"

then $\sum_{k=1}^{\infty} \frac{2}{(-5)^k} = \frac{a}{1-r} = \frac{\frac{-2}{5}}{1+\frac{1}{5}} = \frac{\frac{2}{5}}{\frac{6}{5}} = \frac{-1}{3}$

(*) $S_n = v_1 + v_2 + \cdots + v_{n-1} + v_n$

$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n}\right) + \left(\frac{1}{n} - \frac{1}{n+1}\right)$

$\{S_n\}_{n=1}^{\infty} = \left\{1 - \frac{1}{n+1}\right\}_{n=1}^{\infty}$

$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right) = 1$ 'conv'

$\Rightarrow \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1}\right) = 1$

Then $\sum_{k=1}^{\infty} \left(\frac{2}{(-5)^k} + \frac{1}{k} - \frac{1}{k+1}\right)$ conv.

$= \frac{-1}{3} + 1 = \frac{2}{3}$

9,5 The comparisan. Ratio Root Test:-

Theorem (The comparisan Test (a)):

Let $\sum u_k$ and $\sum v_k$ be series with positive terms and suppose that $u_1 \le v_1$, $u_2 \le v_2 \le \cdots$, $u_k \le v_k \cdots$

① if the bigger series "$\sum v_k$" is Convergent then the smaller series "$\sum u_k$" also "conv"

② if the smaller series "$\sum u_k$" is divergent then the bigger "$\sum v_k$" also "div"

Amazing

Ex: ① $\sum_{k=1}^{\infty} \frac{1}{2k^2+k}$

Sol:- $u_k = \frac{1}{2k^2+k} \quad v_k \; \frac{1}{2k^2} \to \frac{1}{3} < \frac{1}{2}$

Since $\sum_{k=1}^{\infty} \frac{1}{k^2}$ "conv" Test(5) $p=2>1$

$\Rightarrow \sum_{k=1}^{\infty} \frac{1}{2k^2}$ conv. and bigger series

$\Rightarrow \sum_{k=1}^{\infty} \frac{1}{2k^2+k}$ "conv"

② $\sum_{k=2}^{\infty} \frac{1}{\sqrt{k}-1}$

Sol: $u_k = \frac{1}{\sqrt{k}-1} > \frac{1}{\sqrt{k}} = v_k$

$k=2 \quad u_k \; \frac{1}{\sqrt{2}-1} > \frac{1}{\sqrt{2}} = v_k$

Since $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ div $p=\frac{1}{2}<1$ test(5)

$\Rightarrow \sum_{k=2}^{\infty} \frac{1}{\sqrt{k}-2}$ div.

③ $\sum_{k=1}^{\infty} \frac{3k^3-2k^2+4}{k^7-k^3+2}$

Sol: $u_k = \frac{3k^3-2k^2+4}{k^7-k^3+2} < \frac{3k^3}{k^7} = \frac{3}{k^4} \quad v_k$

$\frac{5}{2} < \frac{3}{1}$

since $\sum_{k=1}^{\infty} \frac{1}{k^4}$ "conv" $u=p<1$

$\Rightarrow \sum_{k=1}^{\infty} \frac{3}{k^4}$ conv and bigger

Hence $\sum_{k=1}^{\infty} \frac{3k^3-2k^2+4}{k^7-k^3+2}$ "conv"

Theorem (The limit comparison Test ⑦)

"Let $\sum u_k$ and $\sum v_k$ be series with positive terms and suppose that $\rho = \lim_{k\to\infty} \frac{u_k}{v_k}$ if ρ is finite and $\rho > 0$, then the series both conv, or both div

① $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}+1}$

Sol: $\lim \frac{\frac{1}{\sqrt{k}+1}}{\frac{1}{\sqrt{k}}} = \frac{1}{2}$

Sol: $u_k = \frac{1}{\sqrt{k}+1} < \frac{1}{\sqrt{k}} = v_k$

$\frac{1}{2} < 1$

since $\sum_{k=1}^{\infty} \frac{1}{\sqrt{k}}$ div. $p=\frac{1}{2}<1$

and bigger series

$\rho = \lim_{k\to\infty} \frac{\frac{1}{\sqrt{k}+1}}{\frac{1}{\sqrt{k}}} = \lim_{k\to\infty} \frac{\sqrt{k}}{\sqrt{k}+1}$

$= \lim_{k\to\infty} \frac{\frac{1}{2\sqrt{k}}}{\frac{1}{2\sqrt{k}}} = 1 > 0 \to \rho$

$\Rightarrow \sum_{k=1}^{\infty} \frac{1}{\sqrt{k}+1}$ div.

Amazing

② $\sum_{k=2}^{\infty} \frac{1}{k^2-k}$

Sol: $u_k = \frac{1}{k^2-k} > \frac{1}{k^2} = v_k$

$k=2 \quad \frac{1}{2} > \frac{1}{4}$

since $\sum_{k=1}^{\infty} \frac{1}{k^2}$ conv. $p>1$ and smaller series

$\rho = \lim_{k\to\infty} \frac{\frac{1}{k^2-k}}{\frac{1}{k^2}} = \lim_{k\to\infty} \frac{k^2}{k^2-k}$

$\lim_{k\to\infty} \frac{2k}{2k-1} \Rightarrow \lim_{k\to\infty} \frac{2}{2} = 1 > 0 \to \rho$

$\Rightarrow$ Then $\sum_{k=2}^{\infty} \frac{1}{k^2-k}$ "conv." $\Leftrightarrow$

Theorem (The Ratio Test ⑧)

Let $\sum u_k$ be a series with the positive terms and Suppose that $\rho = \lim_{k\to\infty} \frac{u_{k+1}}{u_k}$

① if $\rho < 1$, then the series converges

② if $\rho > 1$ or $\rho = \infty$, //// /// diverges

③ if $\rho = 1$, the test fails.

Ex: Determine wether the series converge

① $\sum_{k=1}^{\infty} \frac{k}{2^k}$

$k! = k(k-1)(k-2)\dots(2)(1)$

$5! = (5)(4)(3)(2)(1) = 12$

② $\sum_{k=1}^{\infty} \frac{k}{2^k}$

Sol: $u_k = \frac{k}{2^k} \Rightarrow u_{k+1} = \frac{k+1}{2^{k+1}}$

$\frac{u_{k+1}}{k} = \frac{\frac{k+1}{2^{k+1}}}{\frac{k}{2^k}} = \frac{k+1}{2^{k+1}} \cdot \frac{2^k}{k}$

$= \frac{k+1}{2k} \to \rho = \lim_{k\to\infty} \frac{k+1}{2k} = \frac{1}{2} < 1$

"conv"

② $\sum_{k=1}^{\infty} \frac{(2k)!}{4^k} u_k$ $\quad 2(k+1)$

Sol: $u_{k+1} = \frac{(2k+2)!}{4^{k+1}}$

$\frac{u_{k+1}}{u_k} = \frac{(2k+2)!}{4^{k+1}} \cdot \frac{4^k}{(2k)!} =$

$= \frac{(2k+2)(2k+1)(2k)!}{4 \cdot (2k)!}$

$\rho = \lim_{k\to\infty} \frac{1}{4}(4k^2+6k+2) = \lim_{k\to\infty} \frac{1}{4}(4k^2)$

$= (\infty)^2 = \infty \to$ "D.V"

Amazing

③ $\sum_{n=2}^{\infty} \frac{\ln(n)}{e^n} u_n$

$u_{n+1} = \frac{\ln(n+1)}{e^{n+1}}$

$\frac{u_{n+1}}{u_n} = \frac{\ln(n+1)}{e^{n+1}} \cdot \frac{e^n}{\ln(n)}$

$= \frac{1}{e} \cdot \frac{\ln(n+1)}{\ln(n)}$

$\rho = \lim_{n\to\infty} \frac{1}{e} \cdot \frac{\ln(n+1)}{\ln(n)}$

$= \frac{1}{e} \cdot \lim_{n\to\infty} \frac{\frac{1}{n+1}}{\frac{1}{n}} = \frac{1}{e} \cdot \lim_{n\to\infty} \frac{n}{n+1}$

$= \frac{1}{e} \cdot 1 = \frac{1}{e} < 1$ "conv"

④ $\sum_{k=1}^{\infty} \frac{(k!)^2 \cdot 2^k}{(2k+2)!}$

$\frac{u_{k+1}}{u_k} = \frac{((k+1)!)^2 \cdot 2^{k+1}}{(2k+4)!} \cdot \frac{(2k+2)!}{(k!)^2 \cdot 2^k}$

$= \frac{2((k+1)\cdot k!)^2}{(2k+4)(2k+3)(2k+2)!} \cdot \frac{(2k+2)!}{(k+1)^2} = \frac{2(k+1)^2}{(2k+4)(2k+3)}$

$\rho = \lim_{k\to\infty} \frac{2(k^2+2k+1)}{(2k+4)(2k+3)} = \frac{2}{4} = \frac{1}{2} < 1$

"conv"

⑤ $\sum_{k=1}^{\infty} \frac{k^k}{k!}$

Theorem (The Root Test ⑪):
Let $\sum u_k$ be a series with positive terms and suppose that

$\rho = \lim_{k\to\infty} \sqrt[k]{u_k} = \lim_{k\to\infty} (u_k)^{\frac{1}{k}}$

① if $\rho < 1$, then the series converges
② if $\rho > 1$ or $\rho = \infty$, it diverges
③ if $\rho = 1$, the test fails

Ex: ① $\sum_{k=1}^{\infty} \left(\frac{4k+5}{2k+1}\right)^k$

Sol: $u_k = \left(\frac{4k+5}{2k+1}\right)^k$

$(u_k)^{\frac{1}{k}} = \left(\left(\frac{4k+5}{2k+1}\right)^k\right)^{\frac{1}{k}} = \frac{4k+5}{2k+1}$

$\rho = \lim_{k\to\infty} \frac{4k+5}{2k+1} = \frac{4}{2} = 2 > 1$ "div"

② $\sum_{k=1}^{\infty} \left(\frac{k^2+k+1}{100}\right)^{2k} \Rightarrow u_k$

Sol: $(u_k)^{\frac{1}{k}} = \left(\left(\frac{k^2+k+1}{100}\right)^{2k}\right)^{\frac{1}{k}} = \left(\frac{k^2+k+1}{100}\right)^2$

$\rho = \lim_{k\to\infty} \left(\frac{k^2+k+1}{100}\right)^2$

$= \left(\lim_{k\to\infty} \frac{k^2}{100}\right)^2 = \infty^2 = \infty$ "div"

Amazing

③ $\sum_{k=1}^{\infty} \left(\frac{k+5}{k}\right)^{k^2}$

Sol: $\rho = \lim_{k\to\infty} \left(\left(\frac{k+5}{k}\right)^{k^2}\right)^{\frac{1}{k}}$

$= \lim_{k\to\infty} \left(\frac{k+5}{k}\right)^{k} = \left(1+\frac{5}{k}\right)^{k}$

$= e^{\lim_{k\to\infty} \ln\left[\left(1+\frac{5}{k}\right)^{k}\right]} = e^{\lim_{k\to\infty} \left(\frac{\ln\left(1+\frac{5}{k}\right)}{\frac{1}{k}}\right)}$

$\lim_{u\to\infty} e^{\frac{\frac{5}{k^2}}{\frac{k+5}{k}} \cdot k^2}$

$\lim_{u\to\infty} e^{\frac{5k}{k+5}} = e^{\lim_{n\to\infty} \frac{5}{1}} = e^5 > 1$

"div"

9.6 Alternating Series

Def: An Alternating Series has one of the following two forms:

$\sum_{k=1}^{\infty} (-1)^{k+1} \cdot a_k = a_1 - a_2 + a_3 - \cdots$ (1)

$\sum_{k=1}^{\infty} (-1)^{k} \cdot a_k = -a_1 + a_2 - a_3 + \cdots$ (2)

Theorem (Alternating Series test 1):
An alternating series of either form (1) or from (2) converges

if the following cordition are satisfied

① $a_1 \geq a_2 \geq a_3 \geq \cdots$ (Decreasing)

② $\lim_{k\to\infty} a_k = 0$

Amazing

Ex: Test the convergence of the following series:

① $\sum_{k=1}^{\infty} (-1)^k \cdot \frac{1}{k}$ ② $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k}{3^k}$

③ $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot (k+3)}{k(k+1)}$

④ $\sum_{k=1}^{\infty} (-1)^k \cdot \frac{Ln(k)}{k}$

---

① $\sum_{k=1}^{\infty} (-1)^k \cdot \frac{1}{k}$

$a_k = \frac{1}{k} \Rightarrow a_1 = 1 > a_2 = \frac{1}{2} > a_3 = \frac{1}{3} > \cdots$

$\lim_{k\to\infty} \frac{1}{k} = \frac{1}{\infty} = 0 \Rightarrow$ "Conv."

② $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k}{3^k}$

$a_k = \frac{k}{3^k} \Rightarrow a_1 = \frac{1}{3} > a_2 = \frac{2}{9} > a_3 = \frac{1}{9} > \cdots$

$\lim_{k\to\infty} \frac{k}{3^k} = \lim_{k\to\infty} \frac{1}{3^k \cdot Ln3} = \frac{1}{\infty \cdot Ln3} = 0$ "Conv"

③ $\sum_{k=1}^{\infty} \frac{(-1)^{k+1} \cdot (k+3)}{k(k+1)}$

$a_k = \frac{k+3}{k(k+1)} \Rightarrow a_1 = \frac{4}{2} > a_2 = \frac{5}{6} > a_3 = \frac{6}{12} > \cdots$

$\lim_{k\to\infty} \frac{k+3}{k^2+k} = \lim_{k\to\infty} \frac{1}{k} = \frac{1}{\infty} = 0 \Rightarrow$ "Conv"

④ $\sum_{k=1}^{\infty} (-1)^k \cdot \frac{Ln(k)}{k}$

$a_k = \frac{Ln(k)}{k} \Rightarrow a_1 = 0 < a_2 = \frac{Ln2}{2} > \frac{Ln3}{3} > \frac{Ln4}{4} > \cdots$

متناقص ✗

اختبار التقارب المتسلسلة

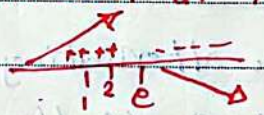
$f(x) = \frac{Lnx}{x} \Rightarrow f'(x) = \frac{x \cdot \frac{1}{x} - Lnx}{x^2}$

$= \frac{1 - Lnx}{x^2} = \frac{0}{1}$

$1 - Lnx = 0 \Rightarrow Lnx = 1 \Rightarrow x = e^1 = e$

f' does not exist at $x = 0$

sign(f')



$f'(1) = 1 > 0$

$f'(e^2) = \frac{1 - Lne^2}{(e^2)^2} = \frac{1-2}{e^4} = \frac{1}{e^4} < 0$

f decreasing on $[e, \infty)$

$\Rightarrow a_k = \frac{Ln(k)}{k}$ decreasing $[3, \infty)$ $(e \leq k)$

$\lim_{k\to\infty} \frac{Lnk}{k} = \lim_{k\to\infty} \frac{\frac{1}{k}}{1} = \frac{1}{\infty} = 0$

"Conv"

Amazing

⑤ $\sum_{k=1}^{\infty} (-1)^k \cdot \frac{k}{3k+1}$

Sol: $\lim_{k\to\infty} \frac{k}{3k+1} = \frac{1}{3} \neq 0$

By divergence test

$$\lim_{k\to\infty} (-1)^k \frac{k}{3k+1} = \begin{cases} \frac{1}{3} & \text{if } k \text{: even} \\ -\frac{1}{3} & \text{if } k \text{: odd} \end{cases}$$

$\neq 0$

$\Rightarrow$ The series "div"

Remark: In alternating series test if $\lim_{k\to\infty} a_k \neq 0$, then by divergence test the given series is divergent

Def: ① A series $\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 + \cdots$

is said to converge absolutely if the series

$\sum_{k=1}^{\infty} |u_k| = |u_1| + |u_2| + \cdots \Rightarrow$ "conv"

② A series $\sum_{k=1}^{\infty} u_k = u_1 + u_2 + u_3 \cdots$

is said to diverge absolutely if the series

$\sum_{k=1}^{\infty} |u_k| = |u_1| + |u_2| + \cdots$ diverges

Def: if a series converge but diverge absolutely, then it's called conditionally convergent

Remark: Every absolutely convergs is converge (But the convers is not true)

Ex: classify the series absolutely convergent, conditionally Convergent or divergent.

① $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$

Sol: $\sum_{k=1}^{\infty} (-1)^{k+1} \cdot \frac{1}{k}$

By alterating series test:

$a_k = \frac{1}{k} \Rightarrow a_1 = 1 > a_2 = \frac{1}{2} > a_3 = \frac{1}{3} > \cdots$

$\lim_{k\to\infty} \frac{1}{k} = 0 \Rightarrow$ "Conv"

$\Rightarrow$

Amazing

*Take the absolute value for all terms in the series.

$1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots=\sum_{k=1}^{\infty}\frac{1}{k}$

OR $\sum_{k=1}^{\infty}\left|(-1)^{k+1}\cdot\frac{1}{k}\right|=\sum_{k=1}^{\infty}\frac{1}{k}$ div "p=1"

$\Rightarrow 1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots$ "div" absolutly

and so, it's Conditionally convergent

ملاحظة: بلتشيك من العينة الطلقة

إذا طلعت "conv" فهي "absolutely convergent"

اما إذا طلعت "div" لازم نلقى الأساس:

إذا كان "conv" يكون "conditionally conv"

إذا كان "div" يكون "absolutely div"

② $1-\frac{1}{2}-\frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}-\cdots$

Sol: Take the absolute value for all terms:

$1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}-\cdots=\sum_{k=0}^{\infty}\frac{1}{2^k}$

$\sum_{k=0}^{\infty}\left(\frac{1}{2}\right)^k$ "geomatric series"

$|r|=\left|\frac{1}{2}\right|=\frac{1}{2}<1 \Rightarrow$ conv.

$\Rightarrow 1-\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\frac{1}{2^4}-\cdots$

(absolutely "conv.")

③ $\sum_{k=1}^{\infty}(-1)^k\frac{k^2+1}{k^3+1}$

Sol: $\sum_{k=1}^{\infty}\left|(-1)^k\cdot\frac{k^2+1}{k^3+1}\right|$

$=\sum_{k=1}^{\infty}1^k\cdot\frac{k^2+1}{k^3+1}$ By comparison test

$u_k=\frac{k^2+1}{k^3+1}\,\frac{k^2}{k^3}=\frac{1}{k}=v_k$

of

$k=1 \rightarrow 1$

$k=2 \rightarrow \frac{5}{9}>\frac{1}{2}$

Since $\sum_{k=1}^{\infty}\frac{1}{k}$ div. (p=1) and Smaller series

$\Rightarrow \sum_{k=1}^{\infty}\left|(-1)^k\cdot\frac{k^2+1}{k^3+1}\right|$ "div"

$\Rightarrow \sum_{k=1}^{\infty}(-1)^k\cdot\frac{k^2+1}{k^3+1}$ absolutely divergent

*By alternating test:-

$a_k=\frac{k^2+1}{k^3+1} \Rightarrow a_1=1>a_2=\frac{5}{9}>a_3=\frac{10}{28}>\cdots$

Then $\lim_{k\to\infty}\frac{k^2+1}{k^3+1}=0$

$\Rightarrow \sum_{k=1}^{\infty}(-1)^k\cdot\frac{k^2+1}{k^3+1}$ "conv."

it's "Conditionally Conv"

Amazing

(#) $\sum_{k=1}^{\infty} (-1)^k \cdot \frac{2^k}{k!}$

Sol: $\sum_{k=1}^{\infty} \left| (-1)^k \cdot \frac{2^k}{k!} \right| = \sum_{k=1}^{\infty} \frac{2^k}{k!}$

By ratio test:-

$\rho = \lim_{k\to\infty} \frac{2^{k+1}}{(k+1)!} \cdot \frac{k!}{2^k}$

$= \lim_{k\to\infty} \frac{2^k \cdot 2^1}{(k+1)\cdot k!} \cdot \frac{k!}{2^k} = \frac{2}{\infty} = 0 < 1$

"Conv"

$\Rightarrow \sum_{k=1}^{\infty} (-1)^k \cdot \frac{2^k}{k!}$ "Abs. conv." and "Conv"

9.7 & 9.8: Taylor & Maclaurin Series

Def: if f has derivatives of all orders at x_0, then

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k = f(x_0) + f'(x_0)\cdot(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3$$

is called Taylor series for f about $x = x_0$

* In special case when $x_0 = 0$, then

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} \cdot x^k = f(0) + f'(0)\cdot x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}\cdot x^3 + \cdots$$

is called Maclaurin Series for f at $x = 0$

Ex: find Taylor series for $\frac{1}{x}$ about $x = 1$?

Sol: let $f(x) = \frac{1}{x}$, $x_0 = 1 \Rightarrow f(1)$

$f'(x) = \frac{-1}{x^2} \rightarrow f'(1) = -1$

$f''(x) = \frac{2}{x^3} \rightarrow f''(1) = 2$

$f'''(x) = \frac{-6}{x^4} \rightarrow f'''(1) = -6$

$f^{(4)}(x) = \frac{24}{x^5} \rightarrow f^{(4)}(1) = 24$

$\frac{1}{x} = 1 - (x-1) + \frac{2}{2!}(x-1)^2 - \frac{6}{3!}(x-1)^3 + \frac{24}{4!}(x-1)^4 \cdots$

$\frac{1}{x} = 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4 \cdots$

$\frac{1}{x} = \sum_{k=0}^{\infty} (-1)^k \cdot (x-1)^k$

Amazing

ex: find Maclaurin series for:

① $\sin x$

$f(x)=\sin x \longrightarrow f(0)=0$

$f'(x)=\cos x \longrightarrow f'(0)=1$

$f''(x)=-\sin x \longrightarrow f''(0)=0$

$f'''(x)=-\cos x \longrightarrow f'''(0)=-1$

$f^{(4)}(x)=\sin x \longrightarrow f^{(4)}(0)=0$

$f^{(5)}(x)=\cos x \longrightarrow f^{(5)}(0)=1$

$f^{(6)}(x)=-\sin x \longrightarrow f^{(6)}(0)=0$

$f^{(7)}(x)=-\cos x \longrightarrow f^{(7)}(0)=-1$

$\sin(x) = 0 + 1\cdot x + 0 + \frac{-1}{3!}x^3 + 0 + \frac{1}{5!}x^5 + 0 + \frac{-1}{7!}x^7 + \cdots$

$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$

$$\sin(x)=\sum_{k=0}^{\infty}(-1)^k\cdot\frac{x^{2k+1}}{(2k+1)!}$$

ex for you

② $\cos x \stackrel{?}{=} \sum_{k=0}^{\infty}(-1)^k\cdot\frac{x^{2k}}{(2k)!}$

③ $\frac{1}{1-x} \stackrel{?}{=} \sum_{k=0}^{\infty}x^k$

④ $e^x \stackrel{?}{=} \sum_{k=0}^{\infty}\frac{x^k}{k!}$

⑤ $\cos(x^2)=\sum_{k=0}^{\infty}(-1)^k\cdot\frac{(x^3)^{2k}}{(2k)!}$

$=\sum_{k=0}^{\infty}(-1)^k\cdot\frac{x^{4k}}{(2k)!}$

⑥ $x^3\cdot e^{(2x+1)} = x^3\cdot\sum_{k=0}^{\infty}\frac{(2x+1)^k}{k!}$

$=\sum_{k=0}^{\infty}x^3\cdot\frac{(2x+1)^k}{k!}$

⑦ $\frac{x^2}{2+x}=x^2\cdot\frac{1}{2\left(1+\frac{x}{2}\right)}$

$=\frac{x^2}{2}\cdot\frac{1}{1-\left(\frac{-x}{2}\right)}=\frac{x^2}{2}\cdot\sum_{k=0}^{\infty}\left(\frac{-x}{2}\right)^k$

$=\sum_{k=0}^{\infty}\frac{x^2}{2}\cdot(-1)^k\cdot\frac{x^k}{2^k}=\sum_{k=0}^{\infty}(-1)^k\cdot\frac{x^{k+2}}{2^{k+1}}$

Def ① $\sum_{k=0}^{\infty}C_k\cdot x^k = C_0+C_1x+C_2x^2+\cdots$

is called a power series in x (about $x=0$)

② $\sum_{k=0}^{\infty}C_k\cdot(x-x_0)^k = C_0+C_1(x-x_0)+C_2\cdot(x-x_0)^2+\cdots$

is called a power series in $(x-x_0)$ (about $x=x_0$)

Amazing

Ex: find the interval of convergence and radies of convergence of the following series:

① $\sum_{k=0}^{\infty} \underbrace{x^k}_{u_k}$

Theorem (The ratio tests for absolutely convergence)

Let $\sum u_k$ be a series, with non-zero terms and suppose $\rho = \lim_{k\to\infty} \left|\frac{u_{k+1}}{u_k}\right|$

① if $\rho < 1$, then the series is absolutely convergent and it's convergent

② if $\rho > 1$ or $\rho = \infty$, then the series diverges.

③ if $\rho = 1$, the test fails

Sol: $u_k = x^k \longrightarrow u_{k+1} = x^{k+1}$

$$\left|\frac{u_{k+1}}{u_k}\right| = \left|\frac{x^{k+1}}{x^k}\right| = |x|$$

$$\rho = \lim_{k\to\infty} |x| = |x| \cdot \lim_{k\to\infty} (1) = |x| < 1$$

$-1 < x < 1$ ⟵ div | conv | div ⟶

-1 $\quad$ 1

$r = \frac{1-(-1)}{2} = 1$ (radius of Conv.)

At $x=1 \Rightarrow \sum_{k=0}^{\infty} (1)^k = \sum_{k=0}^{\infty} (1)$

By divergence test: $\lim_{k\to\infty} (1) = 1 \neq 0$

$\Rightarrow$ div

At $x=-1 \Rightarrow \sum_{k=0}^{\infty} (-1)^k$

By divergence test:

$$\lim_{k\to\infty} (-1)^k = \begin{cases} -1 & \text{if } k: \text{odd} \\ 1 & \text{if } k: \text{even} \end{cases} \neq 0$$

$\Rightarrow \sum_{k=0}^{\infty} (-1)^k$ div

~~interval of c.~~

interval of conv. is $(-1, 1)$

② $\sum_{k=0}^{\infty} \frac{x^k}{k!}$

Sol: $\left|\frac{u_{k+1}}{u_k}\right| = \left|\frac{x^{k+1}}{(k+1)!} \cdot \frac{k!}{x^k}\right|$

$= \left|\frac{x}{(k+1)\cdot k!} \cdot \frac{k!}{1}\right| = \frac{|x|}{|k+1|}$

$= \frac{|x|}{k+1}$

$\rho = \lim_{k\to\infty} \frac{|x|}{k+1} = |x| \cdot \lim_{k\to\infty} \frac{1}{k+1}$

Amazing

$\rho = |x| \cdot (0) = 0 < 1$ conv.

conv. $-\infty$ ∞

The interval of conv. is $R = (-\infty, \infty)$

$r = \frac{\infty - -\infty}{2} = \infty$

③ $\sum_{k=0}^{\infty} k! \cdot x^k$

Sol: $\left|\frac{u_{k+1}}{u_k}\right| = \left|\frac{(k+1)! \cdot x^{k+1}}{k! \cdot x^k}\right|$

$= \left|\frac{(k+1)\cancel{k!} \cdot x}{\cancel{k!}}\right| = |k+1||x| = |x|(k+1)$

$\rho = \lim_{k\to\infty} |x| \cdot (k+1)$

$= |x| \lim_{k\to\infty} (k+1)$

$= |x| \cdot \infty = \infty$ if $x \neq 0$

'div'

div 0 div

$-\infty$ ∞

The series conv. at $x = 0$

$r = 0$

④ $\sum_{k=4}^{\infty} \frac{(x-5)^k}{k^2}$

Sol: $\left|\frac{u_{k+1}}{u_k}\right| = \left|\frac{(x+5)^{k+1}}{(k+1)^2} \cdot \frac{k^2}{(x-5)^k}\right|$

$= \frac{k^2}{(k+1)^2} \cdot |x-5|$

$\rho = \lim_{k\to\infty} \frac{k^2}{(k+1)^2} \cdot |x-5|$

$= |x-5| \lim_{k\to\infty} \frac{k^2}{k^2+2k+1}$

$\rho = |x-5| \cdot (1) = |x-5| < 1$ "conv"

$-1 < x-5 < 1 \Rightarrow 4 < x < 6$

$+5 \quad +5 \quad +5$

div conv. div

$-\infty$ 4 6 ∞

At $x = 6 \Rightarrow \sum_{k=1}^{\infty} \frac{1}{k^2}$ Conv. $p = 2 > 1$

At $x = 4 \Rightarrow \sum_{k=1}^{\infty} (-1)^k \cdot \frac{1}{k^2}$

By alternating test:-

$a_k = \frac{1}{k^2} \Rightarrow a_1 = 1 > a_2 = \frac{1}{4} > a_3 = \frac{1}{9} > \cdots$

$\lim_{k\to\infty} \frac{1}{k^2} = 0 \Rightarrow$ "conv"

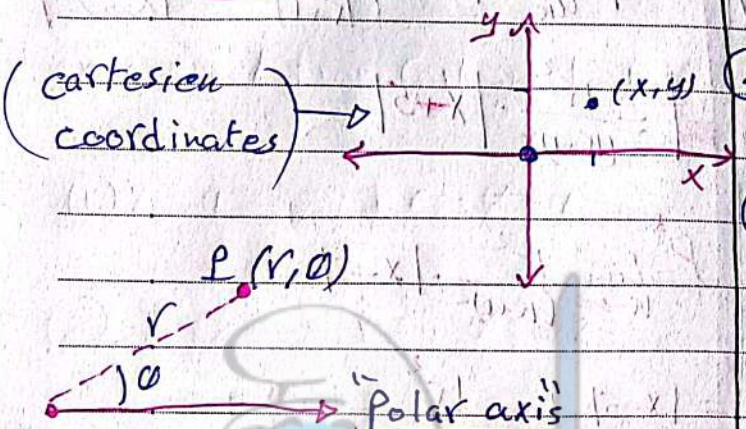
The interval of "conv" is $[4, 6]$

$r = \frac{6-4}{2} = 1$

Amazing

Chapter 10 :-

10,2 Polar Coordinates



Def: Polar coordinate consist of fixed Polar point O it's called Pole (origin) and a ray it's called Polar axis

Remarks: ① Any Point in the Polar Coordinate has the coordinate (r,θ)

② r: Is the distance from O to P

③ θ: is the angle between Polar axis and the Line $\overline{OP}$ (counterclockwise)

④ ~~$(r,\theta)=(r,\theta)$~~

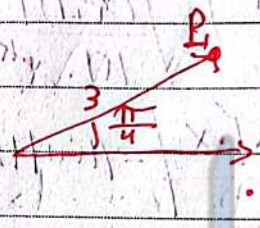
④ $(r,\theta)=(r,\theta\pm 2n\pi)$, $n=1,2,3,4,\dots$

⑤ $(-r,\theta)=(r,\theta+\pi)$

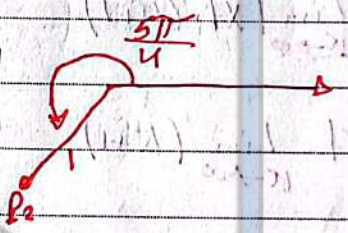
⑥ $(r,-\theta)=(r,2\pi-\theta)$

ex: plot the Points in the Polar Coordinate:

① $P_1\left(3,\frac{\pi}{4}\right)$ (r, θ)



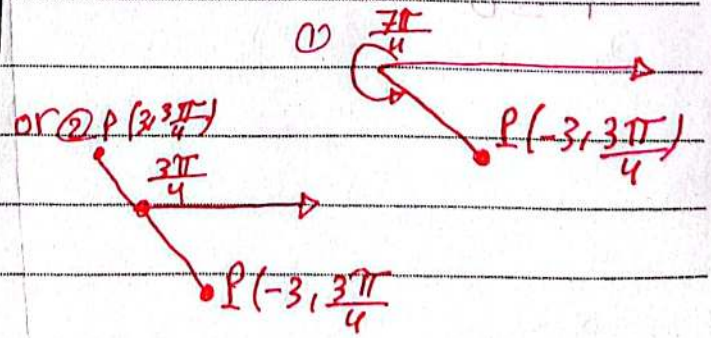
② $P_2\left(1,\frac{5\pi}{4}\right)$



③ $P\left(2,\frac{-2\pi}{3}\right)=P\left(2,2\pi-\frac{2\pi}{3}\right)$

$=P\left(2,\frac{4\pi}{3}\right)$

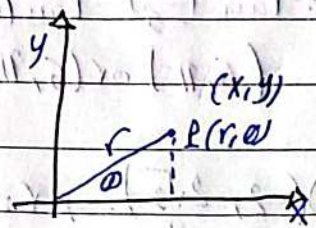
④ $P\left(-3,\frac{3\pi}{4}\right)=P\left(3,\frac{7\pi}{4}\right)$



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* To convert from Polar to ~~rectangular~~ rectangular (Cartesian), we use $\cos\theta = \frac{x}{r}$

$x = r\cos\theta$



$\sin\theta = \frac{y}{r} \Rightarrow y = r\sin\theta$

* // // // Cartesian to Polar

we use: $r^2 = x^2 + y^2 \Rightarrow r = \sqrt{x^2+y^2}$

$\tan\theta = \frac{y}{x} \Rightarrow \theta = \tan^{-1}\frac{y}{x}$

ex: convert from Polar to Cartesian:

① $(6, \frac{2\pi}{3}) \rightarrow r = 6 \ \& \ \theta = \frac{2\pi}{3}$

$x = 6\cdot\cos(\frac{2\pi}{3}) = -6\cdot\cos(\frac{\pi}{3})$

$= -6\cdot\frac{1}{2} = -3$

$y = 6\cdot\sin(\frac{2\pi}{3}) = 6\cdot\sin\frac{\pi}{3}$

$= 6\cdot\frac{\sqrt{3}}{2} = 3\sqrt{3} \Rightarrow (-3, 3\sqrt{3})$

② $(2, \frac{-2\pi}{3}) \Rightarrow x = 2\cdot\cos(\frac{-2\pi}{3})$

$= 2\cdot\cos(\frac{2\pi}{3}) = -1$

$y = 2\cdot\sin(\frac{-2\pi}{3}) = -2\cdot\sin(\frac{2\pi}{3}) = -\sqrt{3}$

$(-1, -\sqrt{3})$

③ $(-4, \frac{-3\pi}{2})$

$x = -4\cdot\cos(\frac{-3\pi}{2}) = -4(0) = 0$

$y = -4\cdot\sin(\frac{-3\pi}{2}) = 4\cdot\sin(\frac{3\pi}{2})$

$= 4(-1) = -4$

$(0, -4)$

④ $r\cos\theta = 3 \Rightarrow x = 3$

⑤ $r = -2\csc\theta \Rightarrow \frac{r}{1} = \frac{-2}{\sin\theta}$

$\Rightarrow r\sin\theta = -2 \Rightarrow y = -2$

⑥ $\theta = \frac{2\pi}{3} \Rightarrow \tan\theta = \tan(\frac{2\pi}{3})$

$\Rightarrow \frac{y}{x} = -\tan(\frac{\pi}{3}) \Rightarrow \frac{y}{x} = \frac{-1}{\sqrt{3}}$

$\Rightarrow \boxed{y = \frac{-1}{\sqrt{3}}x}$

⑦ $r = 3 \Rightarrow \sqrt{x^2+y^2} = 3$

$\Rightarrow x^2 + y^2 = 9$ Circle with Center $(0,0)$ and radius $= 3$

⑧ $r = 4\cos\theta \rightarrow r^2 = 4r\cos\theta$

$\Rightarrow x^2 + y^2 = 4x$

$x^2 - 4x + y^2 = 0 \Rightarrow x^2 - 4x + 4 + y^2 = 4$

$(\frac{-4}{2})^2 = 4 \Rightarrow (x-2)^2 + y^2 = 4$

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⑨ $r=-5\sin\theta \rightarrow r^2=-5r\sin\theta$

$x^2+y^2=-5y$

sol:

ex: convert from Cartesian to Polar:-

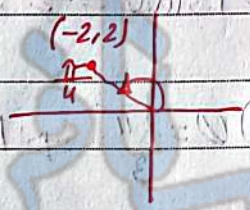
① $x=4 \rightarrow \boxed{r\cos\theta=4}$

② $y=-2 \rightarrow \boxed{r\sin\theta=-2}$

③ $x^2+y^2=3 \rightarrow \boxed{r^2=3}$

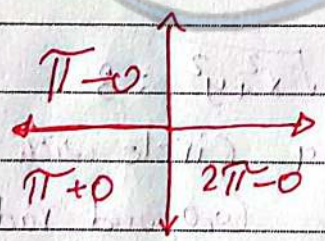
④ $(-2,2)$

sol: $x=-2$ & $y=2$



$r=\sqrt{(-2)^2+2^2}=\sqrt{8}=2\sqrt{2}$

$\theta=\tan^{-1}\left(\frac{2}{-2}\right)=\tan^{-1}(-1)$



$\theta=\frac{3\pi}{4} \rightarrow \left(2\sqrt{2}, \frac{3\pi}{4}\right)$

$\text{or}\left(2\sqrt{2}, \frac{3\pi}{4}\pm 2n\pi\right)$

$n=1,2,3\ldots$

⑤ $(-5,0)$ (x, y) $\rightarrow r=\sqrt{25+0}=5$

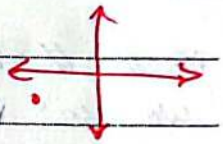
$\theta=\tan^{-1}\left(\frac{0}{-5}\right)=\tan^{-1}(0)=\pi$

$(5,\pi)$ or $(5,\pi\pm 2n\pi)$

⑥ $(-2,-2\sqrt{3})$ (x, y) $\rightarrow r=\sqrt{4+12}=4$

$\theta=\tan^{-1}\left(\frac{-2\sqrt{3}}{-2}\right)=\tan^{-1}(\sqrt{3})$

$\theta=\pi+\frac{\pi}{3}=\frac{4\pi}{3}$

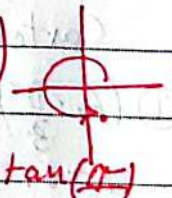


$\left(4,\frac{4\pi}{3}\right)\text{or}\left(4,\frac{4\pi}{3}\pm 2n\pi\right)$

⑦ $(2\sqrt{3},-2) \rightarrow r=4$

$\theta=\tan^{-1}\left(\frac{-2}{2\sqrt{3}}\right)=\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

$=2\pi-\frac{\pi}{6}=\frac{11\pi}{6}$



Amazing